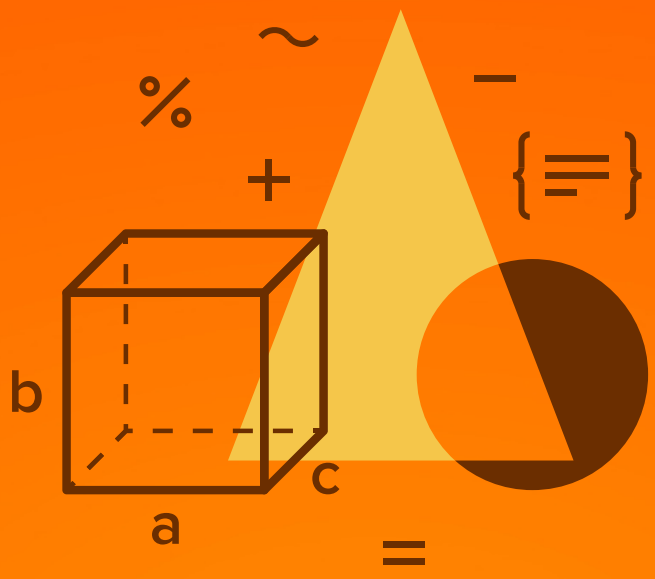
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MATHEMATICS

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Differential Equation & Area Under the Curve

English Medium

EXERCISE (O-1)

Straight Objective Type

1. The differential equation corresponding to the family of curves $y = e^x (ax + b)$ is

(A) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - y = 0$ (B) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = 0$

(C) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0$ (D) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - y = 0$

DE0001

2. Number of values of $m \in N$ for which $y = e^{mx}$ is a solution of the differential equation $D^3y - 3D^2y - 4Dy + 12y = 0$, is

(A) 0 (B) 1 (C) 2 (D) more than 2

DE0002

3. Number of straight lines which satisfy the differential equation $\frac{dy}{dx} + x\left(\frac{dy}{dx}\right)^2 - y = 0$ is:

(A) 1 (B) 2 (C) 3 (D) 4

DE0003

4. The value of the constant 'm' and 'c' for which $y = mx + c$ is a solution of the differential equation $D^2y - 3Dy - 4y = -4x$.

(A) is $m = -1$; $c = 3/4$ (B) is $m = 1$; $c = -3/4$ (C) no such real m, c (D) is $m = 1$; $c = 3/4$

DE0004

5. Consider the two statements

Statement-1: $y = \sin kt$ satisfies the differential equation $y'' + 9y = 0$.

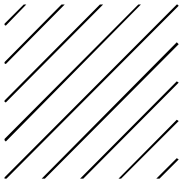
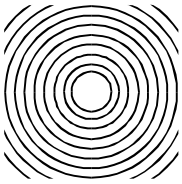
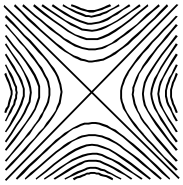
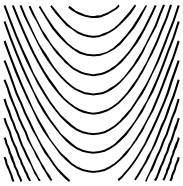
Statement-2: $y = e^{kt}$ satisfy the differential equation $y'' + y' - 6y = 0$

The value of k for which both the statements are correct is

(A) -3 (B) 0 (C) 2 (D) 3

DE0005

6. The general solution of the differential equation $\frac{dy}{dx} = \frac{1-x}{y}$ is a family of curves which looks most like which of the following ?

(A)  (B)  (C)  (D) 

DE0006

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7. If $y = \frac{x}{\ln|cx|}$ (where c is an arbitrary constant) is the general solution of the differential equation $\frac{dy}{dx} = \frac{y}{x} + \phi\left(\frac{x}{y}\right)$ then the function $\phi\left(\frac{x}{y}\right)$ is :

(A) $\frac{x^2}{y^2}$ (B) $-\frac{x^2}{y^2}$ (C) $\frac{y^2}{x^2}$ (D) $-\frac{y^2}{x^2}$

DE0007

8. Water is drained from a vertical cylindrical tank by opening a valve at the base of the tank. It is known that the rate at which the water level drops is proportional to the square root of water depth y, where the constant of proportionality $k > 0$ depends on the acceleration due to gravity and the geometry of the hole. If t is measured in minutes and $k = \frac{1}{15}$ then the time to drain the tank if the water is 4 meter deep to start with is

(A) 30 min (B) 45 min (C) 60 min (D) 80 min

DE0008

9. Spherical rain drop evaporates at a rate proportional to its surface area. The differential equation corresponding to the rate of change of the radius of the rain drop if the constant of proportionality is $K > 0$, is

(A) $\frac{dr}{dt} + K = 0$ (B) $\frac{dr}{dt} - K = 0$ (C) $\frac{dr}{dt} = Kr$ (D) none

DE0009

10. The x-intercept of the tangent to a curve is equal to the ordinate of the point of contact. The equation of the curve through the point (1, 1) is

(A) $ye^{\frac{x}{y}} = e$ (B) $xe^{\frac{x}{y}} = e$ (C) $xe^{\frac{y}{x}} = e$ (D) $ye^{\frac{y}{x}} = e$

DE0010

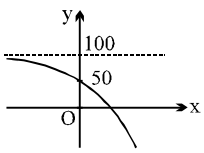
11. A function $f(x)$ satisfying $\int_0^1 f(tx)\,dt = nf(x)$, where $x > 0$, is

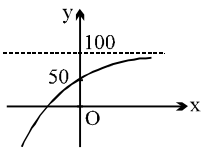
(A) $f(x) = c \cdot x^{\frac{1-n}{n}}$ (B) $f(x) = c \cdot x^{\frac{n}{n-1}}$ (C) $f(x) = c \cdot x^{\frac{1}{n}}$ (D) $f(x) = c \cdot x^{(1-n)}$

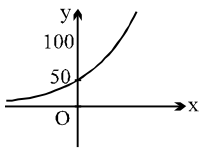
DE0011

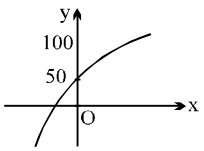
12. Which one of the following curves represents the solution of the initial value problem

$Dy = 100 - y$, where $y(0) = 50$

(A) 

(B) 

(C) 

(D) 

DE0012

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13. The real value of m for which the substitution, $y = u^m$ will transform the differential equation, $2x^4y\frac{dy}{dx} + y^4 = 4x^6$ into a homogeneous equation is :

(A) m = 0 (B) m = 1 (C) m = 3/2 (D) no value of m

DE0013

14. A curve C passes through origin and has the property that at each point (x, y) on it the normal line at that point passes through (1, 0). The equation of a common tangent to the curve C and the parabola $y^2 = 4x$ is

(A) x = 0 (B) y = 0 (C) y = x + 1 (D) x + y + 1 = 0

DE0014

15. A function y = f (x) satisfies $(x + 1) \cdot f'(x) - 2(x^2 + x) f(x) = \frac{e^{x^2}}{(x+1)}, \forall\ x > -1$

If f (0) = 5, then f (x) is

(A) $\left(\frac{3x+5}{x+1}\right).e^{x^2}$ (B) $\left(\frac{6x+5}{x+1}\right).e^{x^2}$ (C) $\left(\frac{6x+5}{(x+1)^2}\right).e^{x^2}$ (D) $\left(\frac{5-6x}{x+1}\right).e^{x^2}$

DE0015

16. The equation to the orthogonal trajectories of the system of parabolas $y = ax^2$ is

(A) $\frac{x^2}{2} + y^2 = c$ (B) $x^2 + \frac{y^2}{2} = c$ (C) $\frac{x^2}{2} - y^2 = c$ (D) $x^2 - \frac{y^2}{2} = c$

DE0016

17. If $\int_a^x t\,y(t)dt = x^2 + y(x)$ then y as a function of x is

(A) $y = 2 - (2 + a^2)e^{\frac{x^2-a^2}{2}}$ (B) $y = 1 - (2 + a^2)e^{\frac{x^2-a^2}{2}}$

(C) $y = 2 - (1 + a^2)e^{\frac{x^2-a^2}{2}}$ (D) none

DE0017

18. If $f(x)$ and g(x) are two differentiable functions on R^+ such that $xf'(x) + g(x) = 0$ and $xg'(x) + f(x) = 0$ for all $x \in R^+$ and $f(1) + g(1) = 4$, then the value of $f''(2).g''(2)$ is -

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{1}{8}$ (D) $\frac{1}{16}$

DE0018

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19. Consider the differential equation $y^2dx+\left(x-\frac{1}{y}\right)dy=0$. If $y(1)=1$, then x is given by :

(A) $1-\frac{1}{y}+\frac{\frac{1}{e^y}}{e}$ (B) $4-\frac{2}{y}-\frac{\frac{1}{e^y}}{e}$ (C) $3-\frac{1}{y}+\frac{\frac{1}{e^y}}{e}$ (D) $1+\frac{1}{y}-\frac{\frac{1}{e^y}}{e}$

DE0101

20. The population p(t) at time t of a certain mouse species satisfies the differential equation $\frac{dp(t)}{dt}=0.5\,p(t)-450$. If $p(0)=850$, then the time at which the population becomes zero is :

(A) ln18 (B) 2 ln18 (C) ln9 (D) $\frac{1}{2}$ ln18

DE0102

21. If $\frac{dy}{dx}=y+3>0$ and $y(0)=2$, then $y(\ln 2)$ is equal to :-

(A) 13 (B) –2 (C) 7 (D) 5

DE0098

22. Let I be the purchase value of an equipment and V(t) be the value after it has been used for t years. The value V(t) depreciates at a rate given by differential equation $\frac{dV(t)}{dt}=-k(T-t)$, where $k>0$ is a constant and T is the total life in years of the equipment. Then the scrap value V(T) of the equipment is :-

(A) $I-\frac{k(T-t)^2}{2}$ (B) e^{-kT} (C) $T^2-\frac{I}{k}$ (D) $I-\frac{kT^2}{2}$

DE0099

23. The curve that passes through the point (2, 3), and has the property that the segment of any tangent to it lying between the coordinate axes is bisected by the point of contact, is given by :

(A) $\left(\frac{x}{2}\right)^2+\left(\frac{y}{3}\right)^2=2$ (B) $2y-3x=0$ (C) $y=\frac{6}{x}$ (D) $x^2+y^2=13$

DE0100

24. The differential equations of all conics whose centre lie at the origin is of order :

(A) 2 (B) 3 (C) 4 (D) none of these

DE0153

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25. The differential equation for all the straight lines which are at a unit distance from the origin is

(A) $\left(y - x\frac{dy}{dx}\right)^2 = 1 - \left(\frac{dy}{dx}\right)^2$ (B) $\left(y + x\frac{dy}{dx}\right)^2 = 1 + \left(\frac{dy}{dx}\right)^2$

(C) $\left(y - x\frac{dy}{dx}\right)^2 = 1 + \left(\frac{dy}{dx}\right)^2$ (D) $\left(y + x\frac{dy}{dx}\right)^2 = 1 - \left(\frac{dy}{dx}\right)^2$ **DE0154**

26. The solution of $y\,dx - x\,dy + 3x^2y^2\,e^{x^3}\,dx = 0$ is

(A) $\frac{x}{y} + e^{x^3} = C$ (B) $\frac{x}{y} - e^{x^3} = 0$ (C) $-\frac{x}{y} + e^{x^3} = C$ (D) $\frac{x}{y} + e^{x^3} = C$

DE0155

27. The solution of the differential equation $(x^2\sin^3 y - y^2\cos x)\,dx + (x^3\cos y\sin^2 y - 2y\sin x)\,dy = 0$ is

(A) $x^3\sin^3 y = 3y^2\sin x + C$ (B) $x^3\sin^3 y + 3y^2\sin x = C$

(C) $x^2\sin^3 y + y^3\sin x = C$ (D) $2x^2\sin y + y^2\sin x = C$ **DE0156**

28. The solution of diffferntial equation $(1 - x^2)\frac{dy}{dx} + xy = ax$ is

(A) $\frac{(y-a)^2 + c^2x^2}{c^2} = 1$ (B) $\frac{(y+a)^2 + c^2x^2}{c^2} = 1$

(C) $\frac{(y+a)^2 - c^2x^2}{c^2} = 1$ (D) $\frac{(y+a)^2 + c^2x^2}{c^2} = -1$ **DE0157**

29. If $\frac{dy}{dx} = e^{-2y}$ and $y = 0$ when $x = 5$, the value of x for $y = 3$ is

(A) e^5 (B) $e^6 + 1$ (C) $\frac{e^6 + 9}{2}$ (D) $\log_e 6$ **DE0158**

30. If $\frac{dy}{dx} = 1 + x + y + xy$ and $y(-1) = 0$, then function y is

(A) $e^{(1-x)^2/2}$ (B) $e^{(1+x)^2/2} - 1$ (C) $\log_e(1 + x) - 1$ (D) $1 + x$ **DE0159**

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EXERCISE (O-2)

Multiple Correct Answer Type

1. The differential equation, $x\dfrac{dy}{dx}+\dfrac{3}{\frac{dy}{dx}}=y^2$:

(A) is of order 1 (B) is of degree 2 (C) is linear (D) is non linear

DE0019

2. Which of the following pair of curves is/are orthogonal ?

(A) $16x^2+y^2=c$ and $y^{16}=kx$ (B) $y=x+ce^{-x}$ and $x+2=y+ke^{-y}$

(C) $y=cx^2$ and $x^2+2y^2=k$ (D) $x^2-y^2=c$ and $xy=k$

where c and k arbitrary constant.

DE0020

3. The function f(x) satisfying the equation, $f^2(x)+4f'(x).f(x)+[f'(x)]^2=0$.

(A) $f(x)=c.e^{(2-\sqrt{3})x}$ (B) $f(x)=c.e^{(2+\sqrt{3})x}$

(C) $f(x)=c.e^{(\sqrt{3}-2)x}$ (D) $f(x)=c.e^{-(2+\sqrt{3})x}$

where c is an arbitrary constant.

DE0021

4. A curve $y=f(x)$ has the property that the perpendicular distance of the origin from the normal at any point P of the curve is equal to the distance of the point P from the x-axis. Then the differential equation of the curve

(A) is homogeneous.

(B) can be converted into linear differential equation with some suitable substitution.

(C) can corresponds to the family of circles touching the x-axis at the origin.

(D) can corresponds to the family of circles touching the y-axis at the origin.

DE0022

5. Consider $g(x)=\begin{cases}\sin x & 0\le x<\dfrac{\pi}{2}\\ \cos x & x\ge\dfrac{\pi}{2}\end{cases}$ and a continuous function $y=f(x)$ satisfies $\dfrac{5dy}{dx}+5y=g(x)$, $f(0)=0$, then -

(A) $f\left(\dfrac{\pi}{4}\right)=\dfrac{e^{-\pi/4}}{10}$ (B) $f\left(\dfrac{\pi}{4}\right)=\dfrac{e^{-\pi/4}-1}{10}$ (C) $f\left(\dfrac{\pi}{2}\right)=\dfrac{e^{-\pi/2}+1}{10}$ (D) $f\left(\dfrac{\pi}{2}\right)=e^{-\pi/2}$

DE0023

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6. A function $y = f(x)$ satisfying the differential equation $\frac{dy}{dx} . \sin x - y\cos x + \frac{\sin^2 x}{x^2} = 0$ is such that, $y \to 0$ as $x \to \infty$ then the statement which is correct is

(A) $\underset{x \to 0}{\text{Limit}}\ f(x) = 1$

(B) $\int_0^{\pi/2} f(x)\,dx$ is less than $\frac{\pi}{2}$

(C) $\int_0^{\pi/2} f(x)\,dx$ is greater than unity

(D) $f(x)$ is an odd function

DE0024

7. If a function $y = f(x)$ satisfies the differential equation $f(x) \cdot \sin 2x - \cos x + (1 + \sin^2 x) f'(x) = 0$ with initial condition $y(0) = 0$. Then-

(A) Range of $f(x)$ is $\left[-\frac{1}{2}, \frac{1}{2}\right]$

(B) $f(1) < f(2)$

(C) $f(1) > f(2)$

(D) $f(x)$ is odd function

DE0025

8. A function $y = f(x)$ satisfies the condition $f'(x) \sin x + f(x) \cos x = 1$, $f(x)$ being bounded when $x \to 0$.

If $I = \int_0^{\pi/2} f(x)\,dx$ then

(A) $\frac{\pi}{2} < I < \frac{\pi^2}{4}$ (B) $\frac{\pi}{4} < I < \frac{\pi^2}{2}$ (C) $f\left(\frac{\pi}{4}\right) < f\left(\frac{\pi}{3}\right)$ (D) $f\left(\frac{\pi}{4}\right) > f\left(\frac{\pi}{3}\right)$

DE0026

9. If $y = f(x)$ is solution of the differential equation, $x^2 \frac{dy}{dx} .\cos\frac{1}{x} - y\sin\frac{1}{x} = -1$, where $y \to -1$ as $x \to \infty$, then

(A) $f(x) = \sin\frac{1}{x} - \cos\frac{1}{x}$

(B) $\lim_{x \to \infty} \left(-f(x)\right)^x = \frac{1}{e}$

(C) $\lim_{x \to \infty} \left(-f(x)\right)^x = e$

(D) $f'(x) + 2xf'(x) + x^2 f''(x) = -\frac{2}{x^2}\sin\frac{1}{x}$

DE0027

10. Consider the differential equation $\frac{dy}{dx} + y\tan x = x\tan x + 1$. Then

(A) The integral curves satisfying the differential equation and given by $y = x + c\cos x$.

(B) The angle at which the integral curves cut the y-axis is $\frac{\pi}{4}$.

(C) Tangents to all the integral curves at their point of intersection with y-axis are parallel.

(D) none of these

DE0028

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11. The differential equation of all circles in a plane must be $\left(y_1=\frac{dy}{dx}, y_2=\frac{d^2y}{dx^2}, ...etc.\right)$

(A) $y_3\left(1+y_1^2\right)-3y_1y_2^2=0$ (B) of order 3 and degree 1

(C) of order 3 and degree 2 (D) $y_3^2\left(1-y_1^2\right)-3y_1y_2^2=0$ **DE0160**

12. Solution of the differential equation $\frac{dy}{dx}+\frac{1+y^2}{\sqrt{1-x^2}}=0$ is

(A) $\tan^{-1}y+\sin^{-1}x=c$ (B) $\tan^{-1}x+\sin^{-1}y=c$

(C) $\tan^{-1}y.\sin^{-1}x=c$ (D) $\cot^{-1}\frac{1}{y}+\cos^{-1}\sqrt{1-x^2}=c$ **DE0161**

13. Correct statement is/are

(A) $f(x,y)=x^2e^{\frac{x}{y}}+\frac{y^3}{x}+y^2\ell n\left(\frac{y}{x}\right)$ is a homogenous function of degree two.

(B) $f(x,y)=\frac{\sin y+x}{\sin 2y+x\cos y}$ is homogenous function of degree one.

(C) $x\sin\left(\frac{y}{x}\right)dy+\left(y\sin\frac{y}{x}-x\right)dx=0$ is a homogenous differential equation.

(D) $f(x,y)=e^{\frac{y}{x}}+\tan\frac{y}{x}$ is homogenous function of degree zero. **DE0162**

14. The solution of $x^2y_1^2+xyy_1-6y^2=0$ are (here $y_1=dy/dx$)

(A) $y=Cx^2$ (B) $x^2y=C$ (C) $\frac{1}{2}\ell ny=C+\log x$ (D) $x^3y=C$ **DE0163**

15. If y(x) satisfies the differential equation $y'-y\tan x=2x\sec x$ and $y(0)=0$, then

(A) $y\left(\frac{\pi}{4}\right)=\frac{\pi^2}{8\sqrt{2}}$ (B) $y'\left(\frac{\pi}{4}\right)=\frac{\pi^2}{18}$ (C) $y\left(\frac{\pi}{4}\right)=\frac{\pi^2}{9}$ (D) $y'\left(\frac{\pi}{3}\right)=\frac{4\pi}{3}+\frac{2\pi^2}{3\sqrt{3}}$

DE0122

16. The solution of $(x+y+1)\,dy=dx$ are

(A) $x+y+2=Ce^y$ (B) $x+y+4==C\log y$

(C) $\log(x+y+2)=Cy$ (D) $\log(x+y+2)=C+y$ **DE0164**

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17. The solution of $\frac{dx}{dy} + y = ye^{(n-1)x}, (n \neq 1)$

(A) $\frac{1}{n-1}\ell n\left(\frac{e^{(n-1)x}-1}{e^{(n-1)x}}\right) = \frac{y^2}{2} + C$ (B) $e^{(1-n)x} = 1 + ce^{(n-1)\frac{y^2}{2}}$

(C) $\ell n\left(1 + ce^{(n-1)\frac{y^2}{2}}\right) + nx + 1 = 0$ (D) $e^{(n-1)x} = ce^{(n-1)+\frac{(n-1)y^2}{2}} + 1$ **DE0165**

18. Correct statement is/are

(A) $f(x, y) = x^2e^{\frac{x}{y}} + \frac{y^3}{x} + y^2\ell n\left(\frac{y}{x}\right)$ is a homogenous function of degree two.

(B) $f(x, y) = \frac{\sin y + x}{\sin 2y + x\cos y}$ is homogenous function of degree one.

(C) $x\sin\left(\frac{y}{x}\right)dy + \left(y\sin\frac{y}{x} - x\right)dx = 0$ is a homogenous differential equation.

(D) $f(x, y) = e^{\frac{y}{x}} + \tan\frac{y}{x}$ is homogenous function of degree zero. **DE0166**

19. The solution of $x^2y_1^2 + xyy_1 - 6y^2 = 0$ are (here $y_1 = dy/dx$)

(A) $y = Cx^2$ (B) $x^2y = C$ (C) $\frac{1}{2}\ell ny = C + \log x$ (D) $x^3y = C$ **DE0167**

20. The solution of differential equation $\left(\frac{dy}{dx}\right)^2 - \frac{dy}{dx}(e^x + e^{-x}) + 1 = 0$ is

(A) $y = e^x + c$ (B) $y = -e^{-x} + c$ (C) $y = 2e^x + 3e^{-x} + c$ (D) $ye^x + 1 = ce^x$

DE0168

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EXERCISE (O-3)

Linked Comprehension Type

Paragraph for Question 1 to 3

Differential equations are solved by reducing them to the exact differential of an expression in x & y i.e., they are reduced to the form $d(f(x, y)) = 0$

e.g. :

$$\frac{xdx + ydy}{\sqrt{x^2 + y^2}} = \frac{ydx - xdy}{x^2}$$

$$\Rightarrow \frac{1}{2}\frac{2xdx + 2ydy}{\sqrt{x^2 + y^2}} = -\frac{xdy - ydx}{x^2} \qquad \Rightarrow d\left(\sqrt{x^2 + y^2}\right) = -d\left(\frac{y}{x}\right)$$

$$\Rightarrow d\left(\sqrt{x^2 + y^2} + \frac{y}{x}\right) = 0$$

$\therefore$ solution is $\sqrt{x^2 + y^2} + \frac{y}{x} = c$.

Use the above method to answer the following question :

1. The general solution of $(2x^3 - xy^2)\,dx + (2y^3 - x^2y)\,dy = 0$ is

(A) $x^4 + x^2y^2 - y^4 = c$ (B) $x^4 - x^2y^2 + y^4 = c$ (C) $x^4 - x^2y^2 - y^4 = c$ (D) $x^4 + x^2y^2 + y^4 = c$

DE0169

2. General solution of the differential equation $\frac{xdy}{x^2 + y^2} + \left(1 - \frac{y}{x^2 + y^2}\right)dx = 0$ is

(A) $x + \tan^{-1}\left(\frac{y}{x}\right) = c$ (B) $x + \tan^{-1}\frac{x}{y} = c$

(C) $x - \tan^{-1}\left(\frac{y}{x}\right) = c$ (D) none of these

DE0170

3. General solution of the differential equation $e^y\,dx + (xe^y - 2y)\,dy = 0$ is

(A) $xe^y - y^2 = c$ (B) $ye^x - x^2 = c$ (C) $ye^y + x = c$ (D) $xe^y - 1 = cy^2$

DE0171

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Paragraph for Question 4 to 6

In order to solve the differential equation of the form $a_0\frac{d^ny}{dx^n}+a_1\frac{d^{n-1}y}{dx^{n-1}}+...+a_ny=0$, where a_0, a_1, a_2 are constants.

We take the auxiliary equation $a_0D^n+a_1D^{n-1}+....+a_n=0$

Find the roots of this equation and then solution of the given differential equation will be as given in the following table.

	Roots of the auxiliary equation	Corresponding complementary function
1.	One real root α_1	$c_1e^{\alpha_1x}$
2.	Two real and different roots α_1 and α_2	$c_1e^{\alpha_1x}+c_2e^{\alpha_2x}$
3.	Two real and equal roots α_1 and α_1	$(c_1+c_2x)e^{\alpha_1x}$
4.	Three real and equal roots $\alpha_1, \alpha_1, \alpha_1$	$(c_1+c_2x+c_3x^2)e^{\alpha_1x}$
5.	One pair of imaginary roots $\alpha\pm i\beta$	$(c_1\cos\beta x+c_2\sin\beta x)e^{\alpha x}$
6.	Two pair of equal imaginary roots $\alpha\pm i\beta$ and $\alpha\pm i\beta$	$[(c_1+c_2x)\cos\beta x+(c_1+c_2x)\sin\beta x]e^{\alpha x}$

Solution of the given differential equation will be y = sum of all the corresponding parts of the complementary functions.

4. Solve $\frac{d^2y}{dx^2}-2\frac{dy}{dx}+y=0.$

(A) $y=(c_1+c_2x)e^x$ (B) $y=(c_1e^x+c_2e^x)$ (C) $y=(c_1x)e^x$ (D) none of these

DE0172

5. Solve $\frac{d^2y}{dx^2}+a^2y=0.$

(A) $y=(c_1\cos ax+c_2\sin ax)e^{ax}$ (B) $y=c_1\cos ax+c_2\sin ax$

(C) $y=c_1e^{ax}+c_2e^{-ax}$ (D) none of these **DE0173**

6. Solve $\frac{d^3y}{dx^3}-6\frac{d^2y}{dx^2}+11\frac{dy}{dx}-6y=0$

(A) $y=(c_1+c_2x+c_3x^2)e^x$ (B) $y=x(c_1e^x+c_2e^{2x}+c_3e^{3x})$

(C) $y=c_1e^x+c_2e^{2x}+c_3e^{3x}$ (D) none of these **DE0174**

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Paragraph for Question 7 and 8

Let f(x) be a differentiable function, satisfying $f(0)=2$, $f'(0)=3$ and $f''(x)=f(x)$

7. If graph of $y=f(x)$ cuts x-axis at $x=\frac{1}{a}\ell n\frac{1}{5}$ then the value of 'a' is **DE0175**

8. If area enclosed by $y=f(x)$ in the second quadrant is A, then the value of [A] is

(Where [.] is greatest integer function) **DE0176**

Matrix Match Type

9. Match the following -

	List-I		**List- II**
(I)	Solution of $y-\frac{xdy}{dx}=y^2+\frac{dy}{dx}$ is	(P)	$xy^2=2y^5+c$
(II)	Solution of $(2x-10y^3)\frac{dy}{dx}+y=0$ is	(Q)	$\sec y=x+1+ce^x$
(III)	Solution of $\sec^2 y\,dy+\tan y\,dx=dx$ is	(R)	$(x+1)(1-y)=cy$
(IV)	Solution of $\sin y\frac{dy}{dx}=\cos y(1-x\cos y)$ is	(S)	$\tan y=1+ce^{-x}$

(A) I → P; II → Q; III → R; IV → S (B) I → R; II → P; III → S; IV → Q

(C) I → P; II → R; III → S; IV → Q (D) I → R; II → S; III → P; IV → Q

DE0177

10. Match the following -

	Column - I		**Column - II**
(A)	$xdy=y(dx+ydy)$, $y(1)=1$ and $y(x_0)=-3$, then $x_0=$	(p)	$\frac{1}{4}$
(B)	If y(t) is solution of $(t+1)\frac{dy}{dt}-ty=1$, $y(0)=-1$, then $y(1)=$	(q)	-15
(C)	$(x^2+y^2)dy=xydx$ and $y(1)=1$ and $y(x_0)=e$, then $x_0=$	(r)	$-\frac{1}{2}$
(D)	$\frac{dy}{dx}+\frac{2y}{x}=0$, $y(1)=1$, then $y(2)=$	(s)	$\sqrt{3}\,e$

DE0178

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EXERCISE (O-4)

Numerical Grid Type

1. If $y = y(x)$ is the solution of the differential equation, $x\frac{dy}{dx} + 2y = x^2$ satisfying $y(1) = 1$, if $y\left(\frac{1}{2}\right) = k$, then find the value of 16k : **DE0179**

2. Let f be a differentiable function such that $f'(x) = 7 - \frac{3}{4}\frac{f(x)}{x}, (x > 0)$ and $f(1) \neq 4$. Then the value of $\lim\limits_{x\to 0^+} x f\left(\frac{1}{x}\right)$ is : **DE0180**

3. Let $y = y(x)$ be the solution of the differential equation, $\left(x^2+1\right)^2\frac{dy}{dx} + 2x\left(x^2+1\right)y = 1$ such that $y(0) = 0$. If $\sqrt{a}\,y(1) = \frac{\pi}{32}$, then the value of '32a' is : **DE0181**

4. If $y = y(x)$ is the solution of the differential equation $\frac{dy}{dx} = (\tan x - y)\sec^2 x$, $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, such that $y(0) = 0$, then $y\left(-\frac{\pi}{4}\right)$ is equal to $(e - k)$, find k : **DE0182**

5. Let $y = y(x)$ be the solution of the differential equation, $\frac{2+\sin x}{y+1}\cdot\frac{dy}{dx} = -\cos x, y > 0, y(0) = 1$. If $y(\pi) = a$ and $\frac{dy}{dx}$ at $x = \pi$ is b, then the value of $(a + b)$ is **DE0183**

6. If $y = y(x)$ is the solution of the differential equation $\frac{5+e^x}{2+y}\cdot\frac{dy}{dx} + e^x = 0$ satisfying $y(0) = 1$, then the value of $|y(\log_e 13)|$ is **DE0184**

7. If $y = \left(\frac{2}{\pi}x - 1\right)\text{cosec}x$ is the solution of the differential equation, $\frac{dy}{dx} + p(x)y = \frac{2}{\pi}\text{cosec}x$, $0 < x < \frac{\pi}{2}$, then find the value of $P\left(\frac{\pi}{4}\right)$. **DE0185**

8. If $y(x)$ satisfies the equation $y'(x) = y(x) + \int_0^1 y\,dx$ & $y(0) = 1$ then value of $y\left(\ell n\frac{11-3e}{2}\right)$ **DE0186**

9. If solution of the differential equation $\frac{dy}{dx} = \frac{1}{x\cos y + \sin 2y}$ is $x = ce^{\sin y} - k(1 + \sin y)$, then $k =$ **DE0187**

10. If the equation of curve passing through (3, 4) and satisfying the differential equation $y\left(\frac{dy}{dx}\right)^2 + (x - y)\frac{dy}{dx} - x = 0$ is $Ax + By + 2 = 0$ then value of $(A - B)$ is **DE0188**

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EXERCISE (JM)

1. At present a firm is manufacturing 2000 items. It is estimated that the rate of change of production P w.r.t. additional number of workers x is given by $\frac{dP}{dx} = 100 - 12\sqrt{x}$. If the firm employs 25 more workers, then the new level of production of items is : **[JEE (Main)-2013]**

(1) 2500 (2) 3000 (3) 3500 (4) 4500

DE0103

2. If the surface area of a sphere of radius r is increasing uniformly at the rate $8cm^2/s$, then the rate of change of its volume is : **[JEE-Main (On line)-2013]**

(1) proportional to r^2 (2) constant (3) proportional to r (4) proportional to $\sqrt{r}$

DE0104

3. Consider the differential equation $\frac{dy}{dx} = \frac{y^3}{2\left(xy^2 - x^2\right)}$:

Statement 1 : The substitution $z = y^2$ transforms the above equation into a first order homogenous differential equation.

Statement 2 : The solution of this differential equation is $y^2 e^{-\frac{y^2}{x}} = C$. **[JEE-Main (On line)-2013]**

(1) Statement 1 is false and statement 2 is true. (2) Both statements are true.

(3) Statement 1 is true and statement 2 is false. (4) Both statements are false.

DE0105

4. If a curve passes through the point $\left(2, \frac{7}{2}\right)$ and has slope $\left(1 - \frac{1}{x^2}\right)$ at any point (x, y) on it, then the ordinate of the point on the curve whose abscissa is – 2 is : **[JEE-Main (On line)-2013]**

(1) $-\frac{5}{2}$ (2) $\frac{5}{2}$ (3) $-\frac{3}{2}$ (4) $\frac{3}{2}$

DE0106

5. The equation of the curve passing through the origin and satisfying the differential equation $(1 + x^2)\frac{dy}{dx} + 2xy = 4x^2$ is : **[JEE-Main (On line)-2013]**

(1) $(1 + x^2)\, y = x^3$ (2) $3(1 + x^2)\, y = 4x^3$ (3) $3(1 + x^2)\, y = 2x^3$ (4) $(1 + x^2)\, y = 3x^3$

DE0107

6. Let the population of rabbits surviving at a time t be governed by the differential equation $\frac{dp(t)}{dt} = \frac{1}{2}p(t) - 200$. If p(0) = 100, then p(t) equals : **[JEE(Main)-2014]**

(1) $400 - 300\, e^{t/2}$ (2) $300 - 200\, e^{-t/2}$ (3) $600 - 500\, e^{t/2}$ (4) $400 - 300\, e^{-t/2}$

DE0108

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7. Let y(x) be the solution of the differential equation $(x \log x)\frac{dy}{dx} + y = 2x \log x, (x \geq 1)$.

Then y(e) is equal to : **[JEE(Main)-2015]**

(1) 2 (2) 2e (3) e (4) 0

DE0109

8. If a curve y = f(x) passes through the point (1, –1) and satisfies the differential equation, $y(1 + xy)\,dx = x\,dy$, then $f\left(-\frac{1}{2}\right)$ is equal to : **[JEE(Main)-2016]**

(1) $\frac{4}{5}$ (2) $-\frac{2}{5}$ (3) $-\frac{4}{5}$ (4) $\frac{2}{5}$

DE0110

9. If $(2+\sin x)\frac{dy}{dx}+(y+1)\cos x=0$ and y(0) = 1, then $y\left(\frac{\pi}{2}\right)$ is equal to :- **[JEE(Main)-2017]**

(1) $\frac{4}{3}$ (2) $\frac{1}{3}$ (3) $-\frac{2}{3}$ (4) $-\frac{1}{3}$

DE0111

10. Let y = y(x) be the solution of the differential equation $\sin x\frac{dy}{dx} + y\cos x = 4x, x \in (0, \pi)$.

If $y\left(\frac{\pi}{2}\right) = 0$, then $y\left(\frac{\pi}{6}\right)$ is equal to : **[JEE(Main)-2018]**

(1) $\frac{-8}{9\sqrt{3}}\pi^2$ (2) $-\frac{8}{9}\pi^2$ (3) $-\frac{4}{9}\pi^2$ (4) $\frac{4}{9\sqrt{3}}\pi^2$

DE0112

11. Let $f : [0, 1] \to R$ be such that $f(xy) = f(x).f(y)$ for all $x,y,\varepsilon[0,1]$, and $f(0) \neq 0$. If y = y(x) satisfies the differential equation, $\frac{dy}{dx} = f(x)$ with y(0) = 1, then $y\left(\frac{1}{4}\right)+y\left(\frac{3}{4}\right)$ is equal to **[JEE(Main)-2019]**

(1) 4 (2) 3 (3) 5 (4) 2

DE0113

12. If $\frac{dy}{dx}+\frac{3}{\cos^2 x}y=\frac{1}{\cos^2 x}, x\in\left(\frac{-\pi}{3},\frac{\pi}{3}\right)$ and $y\left(\frac{\pi}{4}\right)=\frac{4}{3}$, then $y\left(-\frac{\pi}{4}\right)$ equals : **[JEE(Main)-2019]**

(1) $\frac{1}{3}+e^6$ (2) $\frac{1}{3}$ (3) $-\frac{4}{3}$ (4) $\frac{1}{3}+e^3$

DE0114

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13. The curve amongst the family of curves, represented by the differential equation, $(x^2 - y^2)dx + 2xy\,dy = 0$ which passes through (1,1) is : **[JEE(Main)-2019]**

(1) A circle with centre on the y-axis (2) A circle with centre on the x-axis

(3) An ellipse with major axis along the y-axis (4) A hyperbola with transverse axis along the x-axis

DE0115

14. The solution of the differential equation $\frac{dy}{dx} = (x-y)^2$, when $y(1) = 1$, is :- **[JEE(Main)-2019]**

(1) $\log_e\left|\frac{2-y}{2-x}\right| = 2(y-1)$ (2) $\log_e\left|\frac{2-x}{2-y}\right| = x - y$

(3) $-\log_e\left|\frac{1+x-y}{1-x+y}\right| = x + y - 2$ (4) $-\log_e\left|\frac{1-x+y}{1+x-y}\right| = 2(x-1)$

DE0116

15. If a cuve passes through the point (1, –2) and has slope of the tangent at any point (x, y) on it as $\frac{x^2 - 2y}{x}$, then the curve also passes through the point : **[JEE(Main)-2019]**

(1) $\left(-\sqrt{2},1\right)$ (2) $\left(\sqrt{3},0\right)$ (3) (–1, 2) (4) (3, 0)

DE0117

16. Let $y = y(x)$ be the solution curve of the differential equation, $(y^2 - x)\frac{dy}{dx} = 1$, satisfying $y(0) = 1$. This curve intersects the x-axis at a point whose abscissa is : **[JEE(Main)-2020]**

(1) 2 + e (2) 2 (3) 2 – e (4) –e

DE0118

17. If for $x \geq 0$, $y = y(x)$ is the solution of the differential equation **[JEE(Main)-2020]**

$(x+1)dy = ((x+1)^2 + y - 3)dx$, $y(2) = 0$, then $y(3)$ is equal to ———.

DE0119

18. If a curve $y = f(x)$ passes through the point (1, 2) and satisfies $x\frac{dy}{dx} + y = bx^4$, then for what value of b, $\int_1^2 f(x)dx = \frac{62}{5}$? **[JEE(Main)-2021]**

(1) 5 (2) 10 (3) $\frac{62}{5}$ (4) $\frac{31}{5}$ **DE0147**

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19. The population P = P(t) at time 't' of a certain species follows the differential equation $\frac{dP}{dt} = 0.5P - 450$. If P(0) = 850, then the time at which population becomes zero is :

[JEE(Main)-2021]

DE0148

(1) $\log_e 18$ (2) $\log_e 9$ (3) $\frac{1}{2}\log_e 18$ (4) $2\log_e 18$

20. If a curve passes through the origin and the slope of the tangent to it at any point (x, y) is $\frac{x^2 - 4x + y + 8}{x - 2}$, then this curve also passes through the point : **[JEE(Main)-2021]**

(1) (5, 4) (2) (4, 5) (3) (4, 4) (4) (5, 5) **DE0149**

21. A particle is moving in the xy-plane along a curve C passing through the point (3, 3). The tangent to the curve C at the point P meets the x-axis at Q. If the y-axis bisects the segment PQ, then C is a parabola with

[JEE(Main)-2022]

(A) length of latus rectum 3 (B) length of latus rectum 6

(C) focus $\left(\frac{4}{3}, 0\right)$ (D) focus $\left(0, \frac{3}{4}\right)$ **DE0150**

22. The slope of normal at any point (x, y), $x > 0$, $y > 0$ on the curve $y = y(x)$ is given by $\frac{x^2}{xy - x^2y^2 - 1}$. If the curve passes through the point (1, 1), then e.y(e) is equal to **[JEE(Main)-2022]**

(A) $\frac{1 - \tan(1)}{1 + \tan(1)}$ (B) tan(1) (C) 1 (D) $\frac{1 + \tan(1)}{1 - \tan(1)}$ **DE0151**

23. The surface area of a balloon of spherical shape being inflated, increases at a constant rate. If initially, the radius of balloon is 3 units and after 5 seconds,, it becomes 7 units, then its radius after 9 seconds is :

[JEE(Main)-2022]

(A) 9 (B) 10 (C) 11 (D) 12 **DE0152**

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EXERCISE (JA)

1. Let $f:\left[\frac{1}{2},1\right]\to R$ (the set of all real numbers) be a positive, non-constant and differentiable function such that $f'(x) < 2f(x)$ and $f\left(\frac{1}{2}\right)=1$. Then the value of $\int_{1/2}^{1} f(x)dx$ lies in the interval

[JEE(Advanced) 2013, 2M]

(A) $(2e-1, 2e)$ (B) $(e-1, 2e-1)$ (C) $\left(\frac{e-1}{2}, e-1\right)$ (D) $\left(0, \frac{e-1}{2}\right)$

DE0123

2. A curve passes through the point $\left(1, \frac{\pi}{6}\right)$. Let the slope of the curve at each point (x, y) be $\frac{y}{x}+\sec\left(\frac{y}{x}\right), x>0$. Then the equation of the curve is **[JEE(Advanced) 2013, 2M]**

(A) $\sin\left(\frac{y}{x}\right)=\log x+\frac{1}{2}$ (B) $\text{cosec}\left(\frac{y}{x}\right)=\log x+2$

(C) $\sec\left(\frac{2y}{x}\right)=\log x+2$ (D) $\cos\left(\frac{2y}{x}\right)=\log x+\frac{1}{2}$

DE0124

Paragraph for Question 3 and 4

Let $f:[0,1]\to IR$ (the set of all real numbers) be a function. Suppose the function f is twice differentiable, $f(0)=f(1)=0$ and satisfies $f''(x)-2f'(x)+f(x)\geq e^x$, $x\in[0,1]$.

3. If the function $e^{-x}f(x)$ assumes its minimum in the interval $[0,1]$ at $x=\frac{1}{4}$, which of the following is true ?

[JEE(Advanced) 2013, 3, (–1)]

(A) $f'(x)<f(x), \frac{1}{4}<x<\frac{3}{4}$ (B) $f'(x)>f(x), 0<x<\frac{1}{4}$

(C) $f'(x)<f(x), 0<x<\frac{1}{4}$ (D) $f'(x)<f(x), \frac{3}{4}<x<1$

DE0125

4. Which of the following is true for $0<x<1$? **[JEE(Advanced) 2013, 3, (–1)]**

(A) $0<f(x)<\infty$ (B) $-\frac{1}{2}<f(x)<\frac{1}{2}$ (C) $-\frac{1}{4}<f(x)<1$ (D) $-\infty<f(x)<0$

DE0125

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5. The function $y = f(x)$ is the solution of the differential equation $\frac{dy}{dx} + \frac{xy}{x^2-1} = \frac{x^4+2x}{\sqrt{1-x^2}}$ in $(-1, 1)$ satisfying $f(0) = 0$. Then $\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x)dx$ is **[JEE(Advanced)-2014, 3(–1)]**

(A) $\frac{\pi}{3} - \frac{\sqrt{3}}{2}$ (B) $\frac{\pi}{3} - \frac{\sqrt{3}}{4}$ (C) $\frac{\pi}{6} - \frac{\sqrt{3}}{4}$ (D) $\frac{\pi}{6} - \frac{\sqrt{3}}{2}$

DE0126

6. Let y(x) be a solution of the differential equation $(1 + e^x)y' + ye^x = 1$. If $y(0) = 2$, then which of the following statements is(are) true ? **[JEE 2015, 4M, –2M]**

(A) $y(-4) = 0$

(B) $y(-2) = 0$

(C) y(x) has a critical point in the interval $(-1,0)$

(D) y(x) has no critical point in the interval $(-1,0)$

DE0127

7. Consider the family of all circles whose centers lie on the straight line $y = x$. If this family of circles is represented by the differential equation $Py'' + Qy' + 1 = 0$, where P,Q are functions of x,y and y' (here $y' = \frac{dy}{dx}, y'' = \frac{d^2y}{dx^2}$), then which of the following statements is (are) true ?

[JEE 2015, 4M, –2M]

(A) $P = y + x$ (B) $P = y - x$

(C) $P + Q = 1 - x + y + y' + (y')^2$ (D) $P - Q = x + y - y' - (y')^2$

DE0128

8. A solution curve of the differential equation $(x^2 + xy + 4x + 2y + 4)\frac{dy}{dx} - y^2 = 0, x > 0$, passes through the point (1,3). The the solution curve-

(A) intersects $y = x + 2$ exactly at one point

(B) intersects $y = x + 2$ exactly at two points

(C) intersects $y = (x + 2)^2$

(D) does NOT intersect $y = (x + 3)^2$ **[JEE(Advanced)-2016, 4(–2)]**

DE0129

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9. Let $f : (0, \infty) \to \mathbb{R}$ be a differentiable function such that $f'(x) = 2 - \frac{f(x)}{x}$ for all $x \in (0,\infty)$ and $f(1) \neq 1$. Then **[JEE(Advanced)-2016, 4(–2)]**

(A) $\lim\limits_{x\to 0^+} f'\left(\frac{1}{x}\right) = 1$ (B) $\lim\limits_{x\to 0^+} x f\left(\frac{1}{x}\right) = 2$

(C) $\lim\limits_{x\to 0^+} x^2 f'(x) = 0$ (D) $|f(x)| \leq 2$ for all $x \in (0,2)$

DE0130

10. If $y = y(x)$ satisfies the differential equation $8\sqrt{x}\left(\sqrt{9+\sqrt{x}}\right)dy = \left(\sqrt{4+\sqrt{9+\sqrt{x}}}\right)^{-1} dx$, $x > 0$ and $y(0) = \sqrt{7}$, then $y(256) =$ **[JEE(Advanced)-2017, 3(–1)]**

(A) 80 (B) 3 (C) 16 (D) 9

DE0131

11. If $f : \mathbb{R} \to \mathbb{R}$ is a differentiable function such that $f'(x) > 2f(x)$ for all $x \in \mathbb{R}$, and $f(0) = 1$, then **[JEE(Advanced)-2017, 3(–2)]**

(A) $f(x) > e^{2x}$ in $(0,\infty)$ (B) $f(x)$ is decreasing in $(0,\infty)$

(C) $f(x)$ is increasing in $(0,\infty)$ (D) $f'(x) < e^{2x}$ in $(0,\infty)$

DE0132

12. Let $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ be two non-constant differentiable functions. If $f'(x) = (e^{(f(x)-g(x))})g'(x)$ for all $x \in \mathbb{R}$, and $f(1) = g(2) = 1$, then which of the following statement(s) is (are) TRUE ? **[JEE(Advanced)-2018, 4(–2)]**

(A) $f(2) < 1 - \log_e 2$ (B) $f(2) > 1 - \log_e 2$

(C) $g(1) > 1 - \log_e 2$ (D) $g(1) < 1 - \log_e 2$

DE0133

13. Let $f : (0, \pi) \to \mathbb{R}$ be a twice differentiable function such that

$$\lim_{t\to x} \frac{f(x)\sin t - f(t)\sin x}{t - x} = \sin^2 x \text{ for all } x \in (0, \pi).$$

If $f\left(\frac{\pi}{6}\right) = -\frac{\pi}{12}$, then which of the following statement(s) is (are) TRUE ? **[JEE(Advanced)-2018, 4(–2)]**

(A) $f\left(\frac{\pi}{4}\right) = \frac{\pi}{4\sqrt{2}}$ (B) $f(x) < \frac{x^4}{6} - x^2$ for all $x \in (0, \pi)$

(C) There exists $\alpha \in (0, \pi)$ such that $f'(\alpha) = 0$ (D) $f''\left(\frac{\pi}{2}\right) + f\left(\frac{\pi}{2}\right) = 0$

DE0134

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14. Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable function with $f(0) = 0$. If $y = f(x)$ satisfies the differential equation $\frac{dy}{dx} = (2+5y)(5y-2)$, then the value of $\lim\limits_{x\to-\infty} f(x)$ is _____. **[JEE(Advanced)-2018, 3(0)]**

DE0135

15. Let Γ denote a curve $y = y(x)$ which is in the first quadrant and let the point $(1, 0)$ lie on it. Let the tangent to Γ at a point P intersect the y-axis at Y_P. If PY_P has length 1 for each point P on Γ, then which of the following options is/are correct ? **[JEE(Advanced)-2019, 4(–1)]**

(1) $y = \log_e\left(\frac{1+\sqrt{1-x^2}}{x}\right) - \sqrt{1-x^2}$

(2) $xy' - \sqrt{1-x^2} = 0$

(3) $y = -\log_e\left(\frac{1+\sqrt{1-x^2}}{x}\right) + \sqrt{1-x^2}$

(4) $xy' + \sqrt{1-x^2} = 0$

DE0136

16. Let b be a nonzero real number. Suppose $f : \mathbb{R} \to \mathbb{R}$ is a differentiable function such that $f(0)=1$. If the derivative f' of f satisfies the equation $f'(x) = \frac{f(x)}{b^2+x^2}$ for all $x \in \mathbb{R}$, then which of the following statements is/are TRUE ? **[JEE(Advanced)-2020]**

(A) If $b > 0$, then f is an increasing function

(B) If $b < 0$, then f is a decreasing function

(C) $f(x)\, f(-x) = 1$ for all $x \in \mathbb{R}$

(D) $f(x) - f(-x) = 0$ for all $x \in \mathbb{R}$

DE0143

17. For any real numbers α and β, let $y_{\alpha,\beta}(x)$, $x \in \mathbb{R}$, be the solution of the differential equation $\frac{dy}{dx} + \alpha y = xe^{\beta x}$, $y(1) = 1$ **[JEE(Advanced)-2021]**

Let $S = \{y_{\alpha,\beta}(x) : \alpha, \beta \in \mathbb{R}\}$. Then which of the following functions belong(s) to the set S ?

DE0144

(A) $f(x) = \frac{x^2}{2}e^{-x} + \left(e - \frac{1}{2}\right)e^{-x}$

(B) $f(x) = -\frac{x^2}{2}e^{-x} + \left(e + \frac{1}{2}\right)e^{-x}$

(C) $f(x) = \frac{e^x}{2}\left(x - \frac{1}{2}\right) + \left(e - \frac{e^2}{4}\right)e^{-x}$

(D) $f(x) = \frac{e^x}{2}\left(\frac{1}{2} - x\right) + \left(e + \frac{e^2}{4}\right)e^{-x}$

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18. If y(x) is the solution of the differential equation

$$x dy - (y^2 - 4y)dx = 0 \text{ for } x > 0, y(1) = 2,$$

and the slope of the curve $y = y(x)$ is never zero, then the value of $10y\left(\sqrt{2}\right)$ is _____.

[JEE(Advanced)-2022]

DE0145

19. For $x \in \mathbb{R}$, let the function y(x) be the solution of the differential equation

$$\frac{dy}{dx} + 12y = \cos\left(\frac{\pi}{12}x\right), \; y(0) = 0.$$

Then, which of the following statements is/are TRUE ?

(A) y(x) is an increasing function

(B) y(x) is a decreasing function

(C) There exists a real number β such that the line $y = \beta$ intersects the curve $y = y(x)$ at infinitely many points

(D) y(x) is a periodic function

[JEE(Advanced)-2022]

DE0146

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ANSWER KEY

Do yourself - 1

1. one, not defined **2.** two, not defined **3.** one, one

4. order : 2 ; degree : not defined **5.** order : 1 ; degree : not defined

6. order : 3 ; degree : not defined **7.** order : 1 ; degree : not defined

8. order : 1 ; degree : not defined **9.** order : 2 ; degree : 3

10. order : 2 ; degree : not defined

Do yourself - 2

1. **(i)** $y' - y = 2(1 - x)$ **(ii)** $x^2y'' + 2xy' - 2y = 0$ **(iii)** $y''' = 4y'$

2. $xy\frac{d^2y}{dx^2} + x\left(\frac{dy}{dx}\right)^2 - y\frac{dy}{dx} = 0$ **3.** $[1 + (y')^2].\ y''' - 3y'(y'')^2 = 0$

4. $\frac{d^3y}{dx^3} - 6\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 6y = 0$

Do yourself - 3

1. $\ell ny^2 = x + \ell n|x| + k$ **2.** $-\frac{1}{2y^2} = \frac{1}{4}\sqrt{1 + 4x^2} + k$

3. $\sec y = -2\cos x + C$ **4.** $ln^2(\sec x + \tan x) - ln^2(\sec y + \tan y) = c$

5. $lnx\,(1 - y)^2 = c - \frac{1}{2}y^2 - 2y + \frac{1}{2}x^2$ **6.** $\sqrt{x^2 - 1} - \sec^{-1}x + \sqrt{y^2 - 1} = c$

7. $y = c\,(1 - ay)\,(x + a)$ **8.** $y\sin y = x^2 lnx + c$

9. **(a)** $ln\left|\tan\frac{y}{4}\right| = c - 2\sin\frac{x}{2}$, **(b)** $y = e^{\tan(x/2)}$ **10.** 40 **11.** 1/4

Do yourself - 4

1. $\frac{y - 4x + 2}{y - 4x - 2} = ce^{-4x}$ **2.** $2(x - y) = c + \sin 2(x + y)$ **3.** $y = c(1 - ay)(x + a)$

4. $\tan(x + y) - \sec(x + y) = x + c$ **5.** $\frac{-1}{x + y} = x + c$

Do yourself - 5

1. $x + 2y + \ell n|2x - y| + c = 0$ **2.** $\frac{y^2}{2} + 3y - \frac{3x^2}{2} + 5xy = 0$

3. $-3x + 6y - \frac{9}{7}\ell n(14x + 21y - 13) = c'$ **4.** $x^2 - 4xy - y^2 + 10x + 2y = c'$

5. $y^2 - x^2 + xy + 4y - 3x + c = 0$

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Do yourself - 6

1. $x = cye^{xy}$ **2.** $y = cx^2e^{-1/xy}$

Do yourself - 7

1. $\ell n(x^2 + y^2) = 2\tan^{-1}\left(\frac{y}{x}\right) + c$ **2.** $\sqrt{x^2 - y^2} = \sin^{-1}\left(\frac{y}{x}\right) + c$ **3.** $(y+1)^2 = c(x^2 + y^2)$

4. $\sqrt{x^2 + y^2} = \frac{y}{x} + c$

Do yourself - 8

1. 3 **2.** 2/3

3. **(a)** homogeneous **(b)** homogeneous **(c)** not homogeneous
(d) not homogeneous

Do yourself - 9

1. $(3x + y)(x - y) = c_0$ **2.** $y\ell n\left(\frac{y}{x}\right) - y + x\ell nx + cx = 0$ **3.** $x^2y(2x + y) = 3$

4. $c(x-y)^{2/3}(x^2 + xy + y^2)^{1/6} = \exp\left[\frac{1}{\sqrt{3}}\tan^{-1}\frac{x+2y}{x\sqrt{3}}\right]$, where $\exp x \equiv e^x$

Do yourself - 10

1. $x + y - 3 = C(x - y + 1)^3$ **2.** $(x + y - 2) = c(y - x)^3$

3. $\tan^{-1}\frac{y+3}{x+2} + \ell nc\sqrt{(y+3)^2 + (x+2)^2} = 0$ **4.** $x + y + \frac{4}{3} = ce^{3(x-2y)}$

5. $\arctan\frac{2y+1}{2x+1} = \ell nc\sqrt{x^2 + y^2 + x + y + \frac{1}{2}}$

Do yourself - 11

1. $y = \frac{x^4}{2} + 6x^2\ell n|x| - 2x + cx^2$ **2.** $y = 2(x-a)^3 + \frac{c}{(x-a)^3}$ **3.** $2x\,\ell n\,y = \ell n^2\,y + C$

4. $y\sqrt{1+x^2} = c + \frac{1}{2}l n\left[\tan\frac{1}{2}\arctan x\right]$. Another form is $y\sqrt{1+x^2} = c + \frac{1}{2}l n\frac{\sqrt{1+x^2}-1}{x}$

5. $y = c(1 - x^2) + \sqrt{1 - x^2}$ **6.** $y(x-1) = x^2(x^2 - x + c)$

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Do yourself - 12

1. $y = \dfrac{1}{(c-x)e^{3x}}$ **2.** $\dfrac{x^2}{y^2} = -\dfrac{2}{3}x^3\left(\dfrac{2}{3} + \ell nx\right) + c$

3. $\dfrac{1}{y} = -\sin x + ce^x$ **4.** $\dfrac{1}{y^2} = -1 + (c+x)\cot\left(\dfrac{x}{2} + \dfrac{\pi}{4}\right)$

5. $x^3 y^{-3} = 3\sin x + c$ **6.** $y^{-1} e^x = c - x^2$

7. $\left(\dfrac{1}{3} + y\right)\tan^3\dfrac{x}{2} = c + 2\tan\dfrac{x}{2} - x$ **8.** $4(x^2+1)\,y + x^3(1 - 2\,l nx) = cx$

9. $y = cx + x\,l n\tan x$ **10.** $x = ce^{-\arctan y} + \arctan y - 1$

11. $y^2 + x\,l n\,ax = 0$ **12.** $y^2 = 3x^2 - 6x - x^3 + ce^{-x} + 4$

13. $x\,l n\,y = e^x(x-1) + c$ **14.** $\sin y = (e^x + c)(1+x)$

15. $cx^2 + 2xe^{-y} = 1$ **16.** $y = ce^x$; $y = c + \dfrac{x^2}{2}$

17. $y^2 = -1 + (x+1)l n\dfrac{c}{x+1}$ or $x + (x+1)\,l n\dfrac{c}{x+1}$ **18.** $y = \dfrac{1}{x}\tan\left(\ell n|cx|\right)$

19. $e^y = c.\exp(-e^x) + e^x - 1$ **20.** $y = 3\,l n(x^2 + y + 3) + C$

Do yourself - 13

1. **(a)** $y - 2x = K$ **(b)** $y^2 = x + K$ **2.** **(a)** $x^2 + 2y^2 = c$ **(b)** $\sin y = ce^{-x}$

Do yourself - 14

1. $\frac{1}{2}\,\ell n(x^2 + y^2) + y^4 = C$ **2.** $y + x^2 + 1 = Cx$

Do yourself - 15

1. **(a)** $-5e^{-0.2t} + 5$ kg **(b)** $\dfrac{1}{2}(-e^{0.2t} + 1)$ kg/ℓ **2.** $27\dfrac{7}{9}$ minutes

3. $y = 5t\left(1 + \dfrac{50}{50+t}\right)$ gms; $91\dfrac{2}{3}$ gms

Do yourself - 16

1. $\dfrac{x}{y} = x^2 + C$ **2.** $x^2 + y^2 = C$ **3.** $m = m_0 e^{-kt}$ where $k = -\dfrac{1}{t_0} l n\left(1 - \dfrac{\alpha}{100}\right)$

4. $y = kx$ or $xy = c$

5. **(a)** $P = 1000 + 1500e^{-kt}$ where $k = \dfrac{1}{10} l n\left(\dfrac{5}{3}\right)$; **(b)** $T = 10\log_{5/3}(3)$; **(c)** $P = 1000$ as $t \to \infty$

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	C	B	B	A	B	D	C	A	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	B	C	A	B	A	A	B	D	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	D	C	B	C	A	A	A	C	B

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EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,D	A,B,C,D	C,D	A,B,D	A,C	A,B,C	A,B,D	A,B,C	A,B,D	A,B,C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,B	A,D	A,C,D	A,C,D	A,D	A,D	A,B	A,C,D	A,C,D	A,B,D

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9	
Ans.	B	A	A	A	B	C	2.00	1.00	B	
Q.10	A	B	C	D						
	q	r	s	p						

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	49.00	4.00	2.00	2.00	2.00	1	1	4	2	4

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	3	2	3	2	1	1	1	2	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	1	2	4	2	3	3.00	2	4	4
Que.	21	22	23							
Ans.	A	D	A							

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	A	C	D	B	A,C	B,C	A,D	A	B
Que.	11	12	13	14	15	16	17	18	19	
Ans.	A,C	B,C	B,C,D	0.4	1,4	A,C	A,C	8	C	

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EXERCISE (O-1)

Straight Objective Type

1. The area bounded in the first quadrant by the normal at $(1, 2)$ on the curve $y^2 = 4x$, x-axis & the curve is given by :

(A) $\frac{10}{3}$ (B) $\frac{7}{3}$ (C) $\frac{4}{3}$ (D) $\frac{9}{2}$

AU0001

2. Suppose $y = f(x)$ and $y = g(x)$ are two functions whose graphs intersect at the three points (0, 4), (2, 2) and (4, 0) with $f(x) > g(x)$ for $0 < x < 2$ and $f(x) < g(x)$ for $2 < x < 4$. If $\int_0^4 [f(x) - g(x)]dx = 10$ and $\int_2^4 [g(x) - f(x)]dx = 5$, the area between two curves for $0 < x < 2$, is

(A) 5 (B) 10 (C) 15 (D) 20

AU0002

3. Let 'a' be a positive constant number. Consider two curves $C_1 : y = e^x$, $C_2 : y = e^{a-x}$. Let S be the area of the part surrounding by C_1, C_2 and the y-axis, then $\lim_{a \to 0} \frac{S}{a^2}$ equals

(A) 4 (B) 1/2 (C) 0 (D) 1/4

AU0003

4. The area of the region(s) enclosed by the curves $y = x^2$ and $y = \sqrt{|x|}$ is

(A) 1/3 (B) 2/3 (C) 1/6 (D) 1

AU0004

5. Area enclosed by the graph of the function $y = \ln^2 x - 1$ lying in the 4th quadrant is

(A) $\frac{2}{e}$ (B) $\frac{4}{e}$ (C) $2\left(e + \frac{1}{e}\right)$ (D) $4\left(e - \frac{1}{e}\right)$

AU0005

6. The area bounded by the curve $y = f(x)$ (where $f(x) \geq 0$), the co-ordinate axes & the line $x = x_1$ is given by $x_1 \cdot e^{x_1}$. Therefore $f(x)$ equals :

(A) e^x (B) xe^x (C) $xe^x - e^x$ (D) $xe^x + e^x$

AU0006

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7. The slope of the tangent to a curve $y = f(x)$ at $(x, f(x))$ is $2x + 1$. If the curve passes through the point $(1, 2)$ then the area of the region bounded by the curve, the x-axis and the line $x = 1$ is

(A) $\frac{5}{6}$ (B) $\frac{6}{5}$ (C) $\frac{1}{6}$ (D) 1

AU0007

8. The area bounded by the curves $y = x(x-3)^2$ and $y = x$ is (in sq. units) :

(A) 28 (B) 32 (C) 4 (D) 8

AU0008

9. Area of the region enclosed between the curves $x = y^2 - 1$ and $x = |y|\sqrt{1-y^2}$ is

(A) 1 (B) 4/3 (C) 2/3 (D) 2

AU0009

10. The curve $y = ax^2 + bx + c$ passes through the point (1, 2) and its tangent at origin is the line $y = x$. The area bounded by the curve, the ordinate of the curve at minima and the tangent line is

(A) $\frac{1}{24}$ (B) $\frac{1}{12}$ (C) $\frac{1}{8}$ (D) $\frac{1}{6}$

AU0010

11. The area bounded by the curve $y = x e^{-x}$; $xy = 0$ and $x = c$ where c is the x-coordinate of the curve's inflection point, is

(A) $1 - 3e^{-2}$ (B) $1 - 2e^{-2}$ (C) $1 - e^{-2}$ (D) 1

AU0011

12. Area bounded by $12x - y^2 \geq 0$ and $y \geq 2x$ is

(A) 3 (B) 2 (C) 4 (D) 5 **AU0097**

13. If the area bounded between x-axis and the graph of $y = 6x - 3x^2$ between the ordinates $x = 1$ and $x = a$ is 19 square units then 'a' can take the value

(A) 4 or – 2

(B) two values are in (2, 3) and one in (–1, 0)

(C) two values one in (3,4) and one in (–2,–1)

(D) none of these

AU0013

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14. The area of the region enclosed by the curves $y = x$, $x = e$, $y = \frac{1}{x}$ and the positive x-axis is :-

(A) $\frac{3}{2}$ square units (B) $\frac{5}{2}$ square units (C) $\frac{1}{2}$ square units (D) 1 square units

AU0056

15. The area bounded by the curves $y^2 = 4x$ and $x^2 = 4y$ is :-

(A) 0 (B) $\frac{32}{3}$ (C) $\frac{16}{3}$ (D) $\frac{8}{3}$ **AU0057**

16. The area bounded between the parabolas $x^2 = \frac{y}{4}$ and $x^2 = 9y$, and the straight line $y = 2$ is :

(A) $10\sqrt{2}$ (B) $20\sqrt{2}$ (C) $\frac{10\sqrt{2}}{3}$ (D) $\frac{20\sqrt{2}}{3}$

AU0058

17. Let the straight line $x = b$ divide the area enclosed by $y = (1 - x)^2$, $y = 0$ and $x = 0$ into two parts $R_1(0 \le x \le b)$ d and $R_2(b \le x \le 1)$ such that $R_1 - R_2 = \frac{1}{4}$. Then b equals

(A) $\frac{3}{4}$ (B) $\frac{1}{2}$ (C) $\frac{1}{3}$ (D) $\frac{1}{4}$

AU0077

18. Let $f:[-1,2] \to [0,\infty)$ be a continuous function such that $f(x) = f(1-x)$ for all $x \in [-1,2]$. Let $R_1 = \int_{-1}^{2} x f(x)dx$, and R_2 be the area of the region bounded by $y = f(x)$, x=–1, x=2, and the x-axis. Then -

(A) $R_1 = 2R_2$ (B) $R_1 = 3R_2$ (C) $2R_1 = R_2$ (D) $3R_1 = R_2$

AU0078

19. Area enclosed by the curve $y = f(x)$ defined parametrically as $x = \frac{1-t^2}{1+t^2}$, $y = \frac{2t}{1+t^2}$ is equal to

(A) π sq. units (B) $\frac{\pi}{2}$ sq. units (C) $\frac{3\pi}{4}$ sq. units (D) $\frac{3\pi}{2}$ sq. units

AU0098

20. The area bounded by the curve $a^2 y = x^2 (x + a)$ and the x-axis is

(A) $\frac{a^2}{3}$ sq. units (B) $\frac{a^2}{4}$ sq. units (C) $\frac{3a^2}{4}$ sq. units (D) $\frac{a^2}{12}$ sq. units

AU0099

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21. The area inside the parabola $5x^2 - y = 0$ but outside the parabola $2x^2 - y + 9 = 0$ is

(A) $12\sqrt{3}$ sq.units (B) $6\sqrt{3}$ sq.units (C) $8\sqrt{3}$ sq.units (D) $4\sqrt{3}$ sq.units

AU0100

22. The area of the region in 1st quadrant bounded by the y-axis, $y = \frac{x}{4}$, $y = 1 + \sqrt{x}$ and $y = \frac{2}{\sqrt{x}}$ is

(A) $\frac{2}{3}$ sq. units (B) $\frac{8}{3}$ sq. units (C) $\frac{11}{3}$ sq. units (D) $\frac{13}{6}$ sq. units

AU0101

23. Area bounded by $f(x) = \frac{x^2 - 1}{x^2 + 1}$ and the line $y = 1$ is

(A) π sq. units (B) 2π sq. units (C) $\pi/2$ sq. units (D) none of these

AU0102

24. Area bounded by $y = x^3 - x$ and $y = x^2 + x$ is -

(A) $\frac{37}{24}$ (B) $\frac{37}{12}$ (C) $\frac{11}{24}$ (D) $\frac{37}{23}$ **AU0103**

25. The area bounded by the parabola $y^2 = 4ax$, its axis and two ordinates $x = 4$, $x = 9$ is

(A) $4a^2$ (B) $4a^2.4$ (C) $4a^2(9 - 4)$ (D) $\frac{152\sqrt{a}}{3}$ **AU0104**

26. Area bounded by $y = -x^2 + 6x - 5$, $y = -x^2 + 4x - 3$ and $y = 3x - 15$, for $x > 1$, is -

(A) 73 (B) 13/6 (C) 73/6 (D) none of these

AU0105

27. The area bounded by $y = x^2 + 2$ and $y = 2|x| - \cos\pi x$ is equal to -

(A) $\frac{2}{3}$ (B) $\frac{8}{3}$ (C) $\frac{4}{3}$ (D) $\frac{1}{3}$ **AU0106**

28. The area of the smaller portion above x-axis bounded by the curves $y^2 = 8x$ and $\frac{(x-2)^2}{4} + \frac{y^2}{16} = 1$, is -

(A) $4\pi - \frac{32}{3}$ (B) $8\pi - \frac{16}{3}$ (C) $8\pi - \frac{64}{3}$ (D) $2\pi - \frac{16}{3}$ **AU0107**

29. If area bounded by $f(x) = e^x$, $x > 0$, the line $y = x$, y-axis and $y + x = a (a > 1)$ is k then area bounded by $y = \ell nx$, the line $y = x$ between the lines $x + y = 1$ and $x + y = a$ is -

(A) $\frac{2k - 1}{2}$ (B) $\frac{4k - 1}{4}$ (C) a (D) $\frac{4k + 1}{4}$ **AU0108**

30. If area bounded by $y = \frac{a}{4 + x^2}$ $(a > 0)$ and $y = c (c \le 0)$ is an even prime integer, then $[a] + [c]$ is equal to (where [.] denotes greatest integer function) -

(A) 1 (B) 0 (C) –1 (D) 3 **AU0109**

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EXERCISE (O-2)

Multiple Correct Answer Type

1. Let T be the triangle with vertices (0, 0), (0, c^2) and (c, c^2) and let R be the region between $y = cx$ and $y = x^2$ where $c > 0$ then

(A) Area (R) = $\frac{c^3}{6}$

(B) Area of R = $\frac{c^3}{3}$

(C) $\lim\limits_{c\to 0^+} \frac{\text{Area (T)}}{\text{Area (R)}} = 3$

(D) $\lim\limits_{c\to 0^+} \frac{\text{Area (T)}}{\text{Area (R)}} = \frac{3}{2}$

AU0014

2. Suppose f is defined from $R \to [-1, 1]$ as $f(x) = \frac{x^2-1}{x^2+1}$ where R is the set of real number. Then the statement which does not hold is

(A) f is many one onto

(B) f increases for $x > 0$ and decrease for $x < 0$

(C) minimum value is not attained even though f is bounded

(D) the area included by the curve $y = f(x)$ and the line $y = 1$ is π sq. units.

AU0015

3. Consider $f(x) = \begin{cases} \cos x & 0 \le x < \frac{\pi}{2} \\ \left(\frac{\pi}{2} - x\right)^2 & \frac{\pi}{2} \le x < \pi \end{cases}$ such that f is periodic with period π, then

(A) The range of f is $\left[0, \frac{\pi^2}{4}\right)$

(B) f is continuous for all real x, but not differentiable for some real x

(C) f is continuous for all real x

(D) The area bounded by $y = f(x)$ and the X-axis from $x = -n\pi$ to $x = n\pi$ is $2n\left(1 + \frac{\pi^3}{24}\right)$ for a given $n \in N$

AU0016

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4. Which of the following statement(s) is/are True for the function
$f(x) = (x-1)^2(x-2) + 1$ defined on $[0, 2]$?

(A) Range of f is $\left[\frac{23}{27}, 1\right]$.

(B) The coordinates of the turning point of the graph of $y = f(x)$ occur at $(1, 1)$ and $\left(\frac{5}{3}, \frac{23}{27}\right)$.

(C) The value of p for which the equation $f(x) = p$ has 3 distinct solutions lies in interval $\left(\frac{23}{27}, 1\right)$.

(D) The area enclosed by $y = f(x)$, the lines $x = 0$ and $y = 1$ as *x* varies from 0 to 1 is $\frac{7}{12}$.

AU0017

5. If any curve passing through $(-2, 0)$ whose length of sub normal at any point is 4 unit, then -
(A) there are two possible parabolas those touches each other

(B) area bounded by a possible curve and y-axis is $\frac{32}{3}$

(C) there will be a unique such curve

(D) line $x + 2 = 0$ is tangent to the curve

AU0018

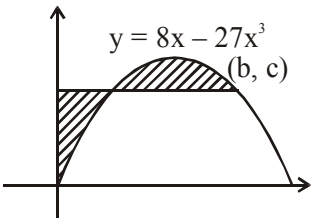
6. Let $f(x) = 2 - |x - 1|$ & $g(x) = (x-1)^2$, then -

(A) area bounded by $f(x)$ & $g(x)$ is $\frac{7}{6}$ (B) area bounded by $f(x)$ & $g(x)$ is $\frac{7}{3}$

(C) area bounded by $f(x)$, $g(x)$ & x-axis is $\frac{5}{3}$ (D) area bounded by $f(x)$, $g(x)$ & x-axis is $\frac{5}{6}$

AU0019

7. The figure shows a horizontal line $y = c$ passing through (b, c) intersecting the curve $y = 8x - 27x^3$. If the shaded areas are equal, then



(A) $b = \frac{1}{9}$ (B) $b = \frac{4}{9}$

(C) $c = \frac{32}{27}$ (D) $c = \frac{23}{27}$

AU0020

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8. If A_1 denotes area of the region bounded by the curves $C_1 : y = (x-1)e^x$, tangent to C_1 at (1,0) & y-axis and A_2 denotes the area of the region bounded by C_1 and co-ordinate axes in fourth quadrant, then -

(A) $A_1 > A_2$ (B) $A_1 < A_2$ (C) $2A_1 + A_2 = 2$ (D) $A_1 + 2A_2 = 4$

AU0021

9. Area bounded by $y = \sec^{-1} x, y = \cot^{-1} x$ and line $x = 1$ is given by -

(A) $\int_1^{\sqrt{\frac{1+\sqrt{5}}{2}}} \left(\cot^{-1} x - \sec^{-1} x\right) dx$

(B) $\int_0^{\alpha} \sec x\, dx + \int_{\alpha}^{\pi/4} \cot x\, dx - \frac{\pi}{4}$, where $\sin\alpha = \cos^2\alpha$

(C) $\int_0^{\alpha} \sec x\, dx + \int_{\alpha}^{\pi/4} \cot x\, dx - \frac{\pi}{4} + 1$, where $\sin\alpha = \cos^2\alpha$

(D) $\int_1^{\frac{1+\sqrt{5}}{2}} \left(\cot^{-1} x - \sec^{-1} x\right) dx$

AU0022

10. Area bounded by the curve $y = \cot x$, $x = \frac{\pi}{4}$ and $y = 0$ is -

(A) $\int_0^{\pi/4} \tan\left(\frac{\pi}{4} - x\right) dx$ (B) $\frac{\pi}{4} - \int_0^1 \tan^{-1} x dx$ (C) $1 - \int_0^1 \tan^{-1} x dx$ (D) $\int_0^{\pi.4} \tan^{-1} x dx$

AU0023

11. Let $f(x) = 2 - |x-1|$ & $g(x) = (x-1)^2$, then -

(A) area bounded by $f(x)$ & $g(x)$ is $\frac{7}{3}$

(B) area bounded by $f(x)$ & $g(x)$ is $\frac{7}{6}$

(C) area bounded by $f(x)$, $g(x)$ & x-axis is $\frac{5}{6}$

(D) area bounded by $f(x)$, $g(x)$ & x-axis is $\frac{5}{3}$ **AU0110**

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12. The function $f(x) = x^{1/3}(x-1)$

(A) has 2 inflection points.

(B) is strictly increasing for $x > 1/4$ and strictly decreasing for $x < 1/4$.

(C) is concave down in $(-1/2,\ 0)$.

(D) Area enclosed by the curve lying in the fourth quadrant is $9/28$. **AU0025**

13. If the curve $y = ax^{1/2} + bx$ passes through the point $(1, 2)$ and lies above the x-axis for $0 \le x \le 9$ and the area enclosed by the curve, the x-axis, and the line $x = 0$ and $x = 4$ is 8 sq. units, then

(A) $a = 1$ (B) $b = 1$ (C) $a = 3$ (D) $b = -1$ **AU0111**

14. Area of the region bounded by the curve $y = e^{x^2}$ and the line $y = e$ is

(A) $2\int_1^e \sqrt{\log_e y}\,dy$ (B) $2e - \int_{-1}^{1} e^{x^2}dx$ (C) $\int_{-1}^{1}\left(e - e^{x^2}\right)dx$ (D) $2\int_0^1 \sqrt{x}e^x dx$

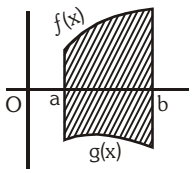
AU0112

15. Which of the following is the possible value/values of c for which the area of the figure bounded by the curves $y = \sin 2x$, the straight lines $x = \frac{\pi}{6}$, $x = c$ and the abscissa axis is equal to 1/2 ?

(A) $-\frac{\pi}{6}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{6}$ (D) none of these

AU0113

16. Which of the following is/are area of shaded region -



(A) $\int_a^b |f(x) - g(x)|\,dx$ (B) $\int_a^b \left(|f(x)| - |g(x)|\right)dx$

(C) $\int_a^b \left(|f(x)| + |g(x)|\right)dx$ (D) $\int_a^b |f(x) + g(x)|\,dx$ **AU0114**

17. Consider $C_1 : y = x^2$ and $C_2 : x^2 + y^2 = 2$. Then,

(A) Area bounded by C_1, C_2 and x-axis in first quadrant will be $\left(\frac{\pi}{4} - \frac{1}{6}\right)$ sq. units

(B) Area bounded by C_1, C_2 and y-axis in second quadrant will be $\left(\frac{7\pi}{4} + \frac{1}{6}\right)$ sq. units

(C) Area bounded by C_1 and C_2 in the major segment will be $\left(\frac{3\pi}{2} - \frac{1}{3}\right)$ sq. units

(D) Area bounded by C_1 and C_2 in the minor segment will be $\left(\frac{\pi}{2} + \frac{1}{3}\right)$ sq. units **AU0115**

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18. Let 'a' be a positive constant number. Consider two curves $C_1 : y = e^x$, $C_2 : y = e^{a-x}$. Let S be the area of the part surrounding by C_1, C_2 and the y-axis, then -

(A) $\lim\limits_{a\to-\infty} S = 1$ (B) $\lim\limits_{a\to0} \frac{S}{a^2} = \frac{1}{4}$

(C) Range of S is $[0,\infty)$ (D) S(a) is neither odd nor even **AU0116**

19. If area enclosed by the graph of function $y = \ell n(x+1)$, y-axis and the line $y = 1$ is A, then choose correct option(s)- (where [.] and {.} denotes greatest integer function and fractional part function respectively)

(A) $\{A\} = e - 2$ (B) $[A] = 1$ (C) $[A] = 0$ (D) $[-A] = -1$

AU0117

20. Area bounded by curve $f(x) = \left|\left|\tan x + \cot x\right| - \left|\tan x - \cot x\right|\right|$ between the lines $x = 0$, $x = \frac{\pi}{2}$ and x-axis is

(A) $\ell n\sqrt{2}$ (B) $2\ell n\sqrt{2}$ (C) $\ell n4$ (D) $\sqrt{2}\ell n2$

AU0118

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EXERCISE (O-3)

Linked Comprehension Type

Paragraph for Question 1 to 3

Consider the curve $x = y^4 - 5y^2 + 4$. It represents following two explicit functions :

$$f : \left[-\frac{9}{4}, 4\right] \to \left[0, \sqrt{\frac{5}{2}}\right]; \quad y = f(x)$$

$$g : \left[-\frac{9}{4}, \infty\right) \to \left[\sqrt{\frac{5}{2}}, \infty\right); \quad y = g(x)$$

Let A_1 be area bounded by $y = f(x)$ & $xy = 0$ as x varies from 0 to 4.

A_2 be area bounded by $y = g(x)$, $x^2 - 16x + 63 = 0$ & $y = 0$.

A_3 be the area bounded by tangent to the curve $y = f(x)$ at $P(-2, \sqrt{2})$ and co-ordinate axes.

On the basis of above information answer the following :

1. A_1 is equal to :-

(A) $\frac{38}{15}$ (B) $\frac{36}{15}$ (C) $\frac{88}{15}$ (D) 3 **AU0119**

2. A_2 is equal to :-

(A) $\int_7^9 \frac{xdx}{2g(x)(2g^2(x) - 5)} + 9g(9) - 7g(7)$ (B) $9g(9) - 7g(7) - \int_7^9 \frac{xdx}{2g(x)(2g^2(x) - 5)}$

(C) $\int_7^9 \frac{xdx}{2g(x)(5 - 2g^2(x))} - 9g(9) + 7g(7)$ (D) none **AU0120**

3. A_3 is equal to :-

(A) $\sqrt{2}$ (B) 2 (C) $\frac{1}{2}$ (D) $\frac{1}{\sqrt{2}}$

AU0121

Paragraph for Question 4 to 6

Given the curves $y = f(x)$ passing through the point (0, 1) and $y = \int_{-\infty}^{x} f(t)dt$ passing through the point $\left(0, \frac{1}{2}\right)$. The tangents drawn to both the curves at the points with equal abscissa intersects on x-axis.

On the basis of above information answer the following :

4. The equation of the curve $y = f(x)$ is -

(A) $\log(x + 2)$ (B) e^{2x} (C) $\frac{3x^2 - x^3}{x^{3/2}}$ (D) $\log\left(\frac{x^2 - x^3}{x^{1/2}}\right)$

AU0122

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5. The area bounded by $y = f(x)$, $y = x$, $x = 0$ and $x = 1$ is -

(A) $\frac{1}{2}(e^2 - 1)$ (B) $\frac{1}{2}(e^2 - 2)$ (C) $(e^2 - 1)$ (D) $(e^2 - 2)$ **AU0123**

6. The value of $\lim_{x \to 0} \frac{(f(x))^3 - 1}{x}$ is -

(A) 3 (B) 6 (C) 4 (D) 2 **AU0124**

Paragraph for Question 7 and 8

Suppose the curve $C : y = x\sqrt{9 - x^2}$ $(x \geq 0)$.

7. The area of the figure bounded by the curve C and the x-axis is - **AU0125**

8. The x-intercept of the tangent drawn to the curve C at $x = \frac{9}{4}$ is - **AU0126**

Matrix Match Type

9.

	List-I	List–II
(I)	The area bounded by the curve $x = 3y^2 - 9$ and the lines $x = 0$, $y = 0$ and $y = 1$ in square units is equal to	(P) 1
(II)	If a curve $f(x) = a\sqrt{x} + bx$, $(f(x) \geq 0 \ \forall \ x \in [0, 9])$ passes through the point (1, 2) and the area bounded by the curve, line $x = 4$ and x-axis is 8 square unit, then $2a + b$ is equal to	(Q) 4
(III)	The area enclosed between the curves $y = \sin^2 x$ and $y = \cos^2 x$ in the interval $0 \leq x \leq \pi$ in square units in equal to	(R) 8
(IV)	The area bounded by the curve $y^2 = 16x$ and line $y = mx$ is $\frac{2}{3}$ square units, then m is equal to	(S) 5

(A) I → S; II → P; III → Q; IV → R (B) I → R; II → S; III → P; IV → Q

(C) I → S; II → R; III → P; IV → Q (D) I → R; II → S; III → Q; IV → P

AU0127

10. Match the areas bounded by the curves in column-I between the ordinates $x = -k$ and $x = k$, $k \in R$ with column-II (where [.] denotes greatest integer function) :

	Column-I	Column-II
(A)	$f(x) = \lvert(-1)^{[x]}\rvert$ and x-axis	(P) $2\lvert k\rvert$
(B)	$f(x) = (-1)^{\text{sgn}[x]}$ and $2x + y = 1$ for $k > 1$	(Q) always a perfect square of integer
(C)	$f(x) = \text{sgn}(x - [x])$ and x-axis	(R) $\frac{k^2}{2}$
(D)	$f(x) = \text{sgn}[x]$ and $y = x$ for $k \in (0, 1)$	(S) $(k - 1)^2$

AU0128

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EXERCISE (O-4)

Numerical Grid Type

1. The area of the region $A=\left[(x,y):0\le y\le x|x|+1 \text{ and } -1\le x\le 1\right]$ in sq. units, is : **AU0129**

2. The area (in sq. units) of the region $A=\{(x,y):\frac{y^2}{2}\le x\le y+4\}$ is : **AU0130**

3. If the area (in sq. units) of the region $\{(x, y) : y^2 \le 4x, x + y \le l, x \ge 0, y \ge O\}$ is $a\sqrt{2}+b$, then $a - b$ is equal to : **AU0131**

4. If the area (in sq. units) bounded by the parabola $y^2 = 4\lambda x$ and the line $y = \lambda x$, $\lambda > 0$, is $\frac{1}{9}$, then λ is equal to : **AU0132**

5. Find the area of the region bounded by curves $f(x) = (x - 4)^2$, $g(x) = 16 - x^2$ and the x-axis. **AU0028**

6. If the area enclosed by the parabolas $y = a - x^2$ and $y = x^2$ is $18\sqrt{2}$ sq. units. Find the value of 'a'. **AU0031**

7. Find the area (in sq. units) enclosed between the curves : $y = \log_e(x+e)$, $x = \log_e(1/y)$ & the x-axis. **AU0034**

8. Let 'c' be the constant number such that $c > 1$. If the least area of the figure given by the line passing through the point $(1, c)$ with gradient '*m*' and the parabola $y = x^2$ is 36 sq. units find the value of $(c^2 + m^2)$. **AU0051**

9. The area (in sq. units) of the region $\{(x, y) \in R^2 | 4x^2 \le y \le 8x + 12)$ is λ, then the value of 3λ is : **AU0133**

10. The area (in sq. units) of the region $\{(x, y) \in R^2 : x^2 \le y \le 3 - 2x\}$, is k, then the value of 3k is : **AU0134**

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EXERCISE (JM)

1. The area (in square units) bounded by the curves $y = \sqrt{x}$, $2y - x + 3 = 0$, x-axis and lying in the first quadrant is : **[JEE (Main)-2013]**

(1) 9 (2) 36 (3) 18 (4) $\frac{27}{4}$

AU0059

2. The area bounded by the curve $y = ln(x)$ and the lines $y = 0$, $y = ln\ (3)$ and $x = 0$ is equal to : **[JEE-Main (On line)-2013]**

(1) $3\ ln\ (3) - 2$ (2) 3 (3) 2 (4) $3\ ln\ (3) + 2$

AU0060

3. The area of the region (in sq. units), in the first quadrant, bounded by the parabola $y = 9x^2$ and the lines $x = 0$, $y = 1$ and $y = 4$, is :- **[JEE-Main (On line)-2013]**

(1) 7/9 (2) 14/3 (3) 14/9 (4) 7/3

AU0061

4. The area under the curve $y = |\cos x - \sin x|$, $0 \le x \le \frac{\pi}{2}$, and above x-axis is : **[JEE-Main (On line)-2013]**

(1) $2\sqrt{2}$ (2) $2\sqrt{2} + 2$ (3) 0 (4) $2\sqrt{2} - 2$

AU0062

5. Let $f : [-2, 3] \to [0, \infty)$ be a continuous function such that $f(1 - x) = f(x)$ for all $x \in [-2, 3]$. If R_1 is the numerical value of the area of the region bounded by $y = f(x)$, $x = -2$, $x = 3$ and the axis of x and $R_2 = \int_{-2}^{3} x\ f(x)\ dx$, then : **[JEE-Main (On line)-2013]**

(1) $2R_1 = 3R_2$ (2) $R_1 = R_2$ (3) $3R_1 = 2R_2$ (4) $R_1 = 2R_2$

AU0063

6. The area of the region described by $A = \{(x, y) : x^2 + y^2 \le 1 \text{ and } y^2 \le 1 - x\}$ is : **[JEE(Main)-2014]**

(1) $\frac{\pi}{2} + \frac{4}{3}$ (2) $\frac{\pi}{2} - \frac{4}{3}$ (3) $\frac{\pi}{2} - \frac{2}{3}$ (4) $\frac{\pi}{2} + \frac{2}{3}$

AU0064

7. The area (in sq.units) of the region $\{(x, y) : y^2 \ge 2x \text{ and } x^2 + y^2 \le 4x, x \ge 0, y \ge 0\}$ is :- **[JEE(Main)-2016]**

(1) $\frac{\pi}{2} - \frac{2\sqrt{2}}{3}$ (2) $\pi - \frac{4}{3}$ (3) $\pi - \frac{8}{3}$ (4) $\pi - \frac{4\sqrt{2}}{3}$

AU0065

8. The area (in sq. units) of the region $\{(x, y\} : x \ge 0, x + y \le 3, x^2 \le 4y \text{ and } y \le 1 + \sqrt{x}\}$ is : **[JEE(Main)-2017]**

(1) $\frac{5}{2}$ (2) $\frac{59}{12}$ (3) $\frac{3}{2}$ (4) $\frac{7}{3}$

AU0066

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9. Let $g(x) = \cos x^2$, $f(x) = \sqrt{x}$ and α, β ($\alpha < \beta$) be the roots of the quadratic equation $18x^2 - 9\pi x + \pi^2 = 0$. Then the area (in sq. units) bounded by the curve $y = (gof)(x)$ and the lines $x = \alpha$, $x = \beta$ and $y = 0$ is - **[JEE(Main)-2018]**

(1) $\frac{1}{2}(\sqrt{3}+1)$ (2) $\frac{1}{2}(\sqrt{3}-\sqrt{2})$ (3) $\frac{1}{2}(\sqrt{2}-1)$ (4) $\frac{1}{2}(\sqrt{3}-1)$

AU0067

10. The area (in sq. units) bounded by the parabola $y = x^2 - 1$, the tangent at the point (2, 3) to it and the y-axis is : **[JEE(Main)-2019]**

(1) $\frac{14}{3}$ (2) $\frac{56}{3}$ (3) $\frac{8}{3}$ (4) $\frac{32}{3}$

AU0068

11. The area of the region $A = \left[(x,y) : 0 \le y \le x|x| + 1 \text{ and } -1 \le x \le 1\right]$ in sq. units, is : **[JEE(Main)-2019]**

(1) $\frac{2}{3}$ (2) $\frac{1}{3}$ (3) 2 (4) $\frac{4}{3}$

AU0069

12. The area (in sq. units) of the region $A = \{(x, y) \in R \times R | 0 \le x \le 3, 0 \le y \le 4, y \le x^2 + 3x\}$ is : **[JEE(Main)-2019]**

(1) $\frac{53}{6}$ (2) $\frac{59}{6}$ (3) 8 (4) $\frac{26}{3}$

AU0070

13. The area (in sq. units) of the region bounded by the curves $y = 2^x$ and $y = |x + 1|$, in the first quadrant is : **[JEE(Main)-2019]**

(1) $\frac{3}{2} - \frac{1}{\log_e 2}$ (2) $\frac{1}{2}$ (3) $\log_e 2 + \frac{3}{2}$ (4) $\frac{3}{2}$

AU0071

14. If the area (in sq. units) of the region $\{(x, y) : y^2 \le 4x, x + y \le 1, x \ge 0, y \ge O\}$ is $a\sqrt{2} + b$, then $a - b$ is equal to : **[JEE(Main)-2019]**

(1) $\frac{8}{3}$ (2) $\frac{10}{3}$ (3) 6 (4) $-\frac{2}{3}$

AU0072

15. The area (in sq. units) of the region $\{(x, y) \in R^2 | 4x^2 \le y \le 8x + 12)$ is : **[JEE(Main)-2020]**

(1) $\frac{127}{3}$ (2) $\frac{125}{3}$ (3) $\frac{124}{3}$ (4) $\frac{128}{3}$

AU0073

16. The area of the region, enclosed by the circle $x^2 + y^2 = 2$ which is not common to the region bounded by the parabola $y^2 = x$ and the straight line $y = x$, is : **[JEE(Main)-2020]**

(1) $\frac{1}{3}(12\pi - 1)$ (2) $\frac{1}{6}(12\pi - 1)$ (3) $\frac{1}{6}(24\pi - 1)$ (4) $\frac{1}{3}(6\pi - 1)$

AU0074

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17. The area (in sq. units) of the region $\{(x, y) \in R^2 : x^2 \le y \le 3 - 2x\}$, is **[JEE(Main)-2020]**

(1) $\frac{29}{3}$ (2) $\frac{31}{3}$ (3) $\frac{34}{3}$ (4) $\frac{32}{3}$

AU0075

18. Given : $f(x) = \begin{cases} x & , \ 0 \le x < \frac{1}{2} \\ \frac{1}{2} & , \ x = \frac{1}{2} \\ 1-x & , \ \frac{1}{2} < x \le 1 \end{cases}$ and $g(x) = \left(x - \frac{1}{2}\right)^2$, $x \in R$. Then the area (in sq. units) of the region bounded by the curves, $y = f(x)$ and $y = g(x)$ between the lines, $2x = 1$ and $2x = \sqrt{3}$, is :

[JEE(Main)-2020]

(1) $\frac{1}{3} + \frac{\sqrt{3}}{4}$ (2) $\frac{\sqrt{3}}{4} - \frac{1}{3}$ (3) $\frac{1}{2} + \frac{\sqrt{3}}{4}$ (4) $\frac{1}{2} - \frac{\sqrt{3}}{4}$

AU0076

19. The area of the region : $R = \{(x, y) : 5x^2 \le y \le 2x^2 + 9\}$ is : **[JEE(Main)-2021]**

(1) $11\sqrt{3}$ square units (2) $12\sqrt{3}$ square units (3) $9\sqrt{3}$ square units (4) $6\sqrt{3}$ square units

AU0091

20. The graphs of sine and cosine functions, intersect each other at a number of points and between two consecutive points of intersection, the two graphs enclose the same area A. Then A^4 is equal to _____.

[JEE(Main)-2021]

AU0092

21. Let A_1 be the area of the region bounded by the curves $y = \sin x$, $y = \cos x$ and y-axis in the first quadrant. Also, let A_2 be the area of the region bounded by the curves $y = \sin x$, $y = \cos x$, x-axis and $x = \frac{\pi}{2}$ in the first quadrant. Then, **[JEE(Main)-2021]**

(1) $A_1 : A_2 = 1 : \sqrt{2}$ and $A_1 + A_2 = 1$ (2) $A_1 = A_2$ and $A_1 + A_2 = \sqrt{2}$

(3) $2A_1 = A_2$ and $A_1 + A_2 = 1 + \sqrt{2}$ (4) $A_1 : A_2 = 1 : 2$ and $A_1 + A_2 = 1$ **AU0093**

22. Let S be the region bounded by the curves $y = x^3$ and $y^2 = x$. The curve $y = 2|x|$ divides S into two regions of areas R_1 and R_2. If $\max\{R_1, R_2\} = R_2$, then $\frac{R_2}{R_1}$ is equal to ____. **[JEE(Main)-2022]**

AU0094

23. The area of the region enclosed between the parabolas $y^2 = 2x - 1$ and $y^2 = 4x - 3$ is

(A) $\frac{1}{3}$ (B) $\frac{1}{6}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$ **AU0095**

24. The area bounded by the curve $y = |x^2 - 9|$ and the line $y = 3$ is :

(A) $4\left(2\sqrt{3} + \sqrt{6} - 4\right)$ (B) $4\left(4\sqrt{3} + \sqrt{6} - 4\right)$ (C) $8\left(4\sqrt{3} + 3\sqrt{6} - 9\right)$ (D) $8\left(4\sqrt{3} + 2\sqrt{6} - 9\right)$

AU0096

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EXERCISE (JA)

1. The area enclosed by the curve $y = \sin x + \cos x$ and $y = |\cos x - \sin x|$ over the interval $\left[0, \frac{\pi}{2}\right]$ is

[JEE(Advanced) 2013, 2M]

(A) $4(\sqrt{2}-1)$ (B) $2\sqrt{2}(\sqrt{2}-1)$ (C) $2(\sqrt{2}+1)$ (D) $2\sqrt{2}(\sqrt{2}+1)$

AU0079

2. Let $F(x) = \int_{x}^{x^2+\frac{\pi}{6}} 2\cos^2 t\,dt$ for all $x \in \mathbb{R}$ and $f : \left[0, \frac{1}{2}\right] \to [0, \infty)$ be a continuous function. For $a \in \left[0, \frac{1}{2}\right]$, if F'(a) + 2 is the area of the region bounded by x = 0, y = 0, y = $f(x)$ and x = a, then $f(0)$ is

[JEE 2015, 4M, –0M]

AU0080

3. If the line $x = \alpha$ divides the area of region $R = \{(x, y) \in \mathbb{R}^2 : x^3 \le y \le x, 0 \le x \le 1\}$ into two equal parts, then **[JEE(Advanced)-2017, 3(–2)]**

(A) $\frac{1}{2} < \alpha < 1$ (B) $\alpha^4 + 4\alpha^2 - 1 = 0$ (C) $0 < \alpha \le \frac{1}{2}$ (D) $2\alpha^4 - 4\alpha^2 + 1 = 0$

AU0081

4. Let $f : [0, \infty) \to \mathbb{R}$ be a continuous function such that $f(x) = 1 - 2x + \int_0^x e^{x-t} f(t)\,dt$

for all $x \in [0, \infty)$. Then, which of the following statement(s) is (are) TRUE ?

(A) The curve y = $f(x)$ passes through the point (1, 2) **[JEE(Advanced)-2018, 4(–2)]**

(B) The curve y = $f(x)$ passes through the point (2, –1)

(C) The area of the region $\{(x, y) \in [0, 1] \times \mathbb{R} : f(x) \le y \le \sqrt{1-x^2}\}$ is $\frac{\pi - 2}{4}$

(D) The area of the region $\{(x, y) \in [0, 1] \times \mathbb{R} : f(x) \le y \le \sqrt{1-x^2}\}$ is $\frac{\pi - 1}{4}$

AU0082

5. A farmer F_1 has a land in the shape of a triangle with vertices at P(0, 0), Q(1, 1) and R(2, 0). From this land, a neighbouring farmer F_2 takes away the region which lies between the side PQ and a curve of the form $y = x^n (n > 1)$. If the area of the region taken away by the farmer F_2 is exactly 30% of the area of ΔPQR, then the value of n is _____ **[JEE(Advanced)-2018, 3(0)]**

AU0083

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6. The area of the region $\{(x, y) : xy \le 8, 1 \le y \le x^2\}$ is **[JEE(Advanced)-2019, 3(–1)]**

(1) $8\log_e 2 - \frac{14}{3}$ (2) $16\log_e 2 - \frac{14}{3}$ (3) $16\log_e 2 - 6$ (4) $8\log_e 2 - \frac{7}{3}$

AU0084

7. Let the functions $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ be defined by

$f(x) = e^{x-1} - e^{-|x-1|}$ and $g(x) = \frac{1}{2}\left(e^{x-1} + e^{1-x}\right)$.

Then the area of the region in the first quadrant bounded by the curves $y = f(x)$, $y = g(x)$ and $x = 0$ is **[JEE(Advanced)-2020]**

(A) $\left(2-\sqrt{3}\right) + \frac{1}{2}(e - e^{-1})$ (B) $\left(2+\sqrt{3}\right) + \frac{1}{2}(e - e^{-1})$

(C) $\left(2-\sqrt{3}\right) + \frac{1}{2}(e + e^{-1})$ (D) $\left(2+\sqrt{3}\right) + \frac{1}{2}(e + e^{-1})$

AU0088

8. The area of the region $\left\{(x, y) : 0 \le x \le \frac{9}{4}, \quad 0 \le y \le 1, \quad x \ge 3y, \quad x + y \ge 2\right\}$ is

[JEE(Advanced)-2021]

(A) $\frac{11}{32}$ (B) $\frac{35}{96}$ (C) $\frac{37}{96}$ (D) $\frac{13}{32}$

AU0089

9. Consider the functions $f, g : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = x^2 + \frac{5}{12} \text{ and } g(x) = \begin{cases} 2\left(1 - \frac{4|x|}{3}\right), & |x| \le \frac{3}{4}, \\ 0, & |x| > \frac{3}{4}. \end{cases}$$

If α is the area of the region

$$\left\{(x, y) \in \mathbb{R} \times \mathbb{R} : |x| \le \frac{3}{4}, 0 \le y \le \min\{f(x), g(x)\}\right\},$$

then the value of 9α is _____. **[JEE(Advanced)-2022]**

AU0090

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ANSWER KEY

Do yourself - 1

1. $\frac{25}{3}$ sq. units **2.** $\frac{\pi}{4}$ sq. units. **3.** 8 sq. units.

4. $\frac{3}{8}$ sq.units **5.** $\frac{117}{3}$ **6.** **4** **7.** $2+e^2-\frac{3}{e^2}$ **8.** $\frac{56}{3}$

Do yourself - 2

1. $\frac{1}{6}$ sq.units **2.** 4 sq. units **3.** $\frac{4-\sqrt{2}}{\log 2}-\frac{5}{2}\log 2+\frac{3}{2}$ sq. units **4.** 5/6 sq. units

5. 23/6 sq. units **6.** $\pi-\tan^{-1}\frac{2\sqrt{2}}{3\pi}$; $\pi-\tan^{-1}\frac{4\sqrt{2}}{3\pi}$

7. $\frac{3\pi+2}{\pi-2}$ **8.** $\sqrt{2}-1$

Do yourself - 3

1. 2π sq. units **2.** e **3.** $C=-1$ or $\left(8-\sqrt{17}\right)^{1/3}$

4. $\frac{1}{3}+\ell n\left(\frac{\sqrt{3}}{2}\right)$ sq. units **5.** $\sqrt{3}$ **6.** $\frac{4-3\ell n3}{2}$

Do yourself - 4

1. $\frac{\pi}{4}$ **2.** $\frac{1}{2}(1-e^{-1/2})$ **3.** **(i)** $m=1$, **(ii)** $m=\infty$; $A_{min}=4/3$

4. $a=-3/4$ **5.** $k=2, A_{min}=\frac{32}{3}$

Do yourself - 5

1. 60 sq. units **2.** e^e-e-1 **3.** 9/2 **4.** 22

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	C	D	B	B	D	A	D	D	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	A	C	A	C	D	B	C	A	D
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	C	B	B	D	C	B	D	B	A

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EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,C	A,C,D	A,D	B,C,D	A,B,D	B,C	B,C	B,C	A,B	A,B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,D	A,B,C,D	C,D	A,B,C,D	A,B	A,C	A,C,D	A,B,C,D	A,C,D	C

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9	
Ans.	A	B	D	B	B	B	9.00	10.12 or 10.13	B	
Q.10	A	B	C	D						
	P	S	P	R						

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	18	6	24	64	9.00	2.00	104.00	128.00	32.00

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	3	3	4	4	1	3	1	4	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	2	1	3	4	2	4	2	2	64
Que.	21	22	23	24						
Ans.	1	19	A	D						

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	
Ans.	B	3	A,D	B,C	4	2	A	A	6	

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